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EXAMINATION
INTELLIGENCE
REPORT

FULL EXAM ANALYSIS REPORT

DETAILED PERFORMANCE • QUESTION ANALYSIS • LEARNING INSIGHTS

NEET Mock Paper Set - 2

Combined (Physics, Chemistry, Biology) | All Subjects

58.1%
Score

Marks Obtained
418 / 720

113 Correct | 34 Incorrect | 33 Skipped



Student	**AAHANA PATTANAIK**
Roll Number	**260201**
Exam Date	**07 Sept 2026**
Total Marks	**720**

Subject Analysis | Chapter Mastery | Question-wise Review | Error & Learning Insights | Revision Plan & Next Steps

DISCIPLINE
CREATIVITY
PROGRESS

ABRIGHTER
TOMORROW
TOGETHER

01 PERFORMANCE INTELLIGENCE

Performance intelligence

Response composition

180 questions in the paper



- Correct 113
- Incorrect 34
- Skipped 33

Mistake distribution

Count by published classification

Calculation Error	6
Knowledge Gap	16
Formula Error	3
Conceptual Error	9

Source classifications; rows are measured independently.

Subject-wise performance

Accuracy by subject

Physics	35.6%
Chemistry	46.7%
Biology	84.4%

Physics 51/180 Chemistry 75/180 Biology 292/360

Important performance signals

GRADE	RANK	BATCH RANK
C	7	7
COHORT SIZE	**OVERALL SPEED**	**CONFIDENCE INDEX**
55	1.03	76.9
READINESS SCORE	**HEALTH SCORE**	**NEGATIVE MARKS**
58.1	58.1	34
GUESS ATTEMPTS	**ATTEMPT RATIO**	**TOTAL QUESTIONS**
0	81.7%	180
OVERALL ACCURACY	**BADGE**	
76.87%	Completed	

02 MASTERY MAP

Subject and chapter mastery

Subject marks, accuracy and classification

SUBJECT	MARKS	ACCURACY	RANK	LEVEL	STRONG AREAS	WEAK AREAS
Physics	51 / 180	35.6%	14	Medium	Position and Displacement · Scalars and Vectors · Average and Instantaneous Velocity	Position and Displacement · Average and Instantaneous Velocity · Acceleration
Chemistry	75 / 180	46.7%	6	Medium	Dalton's Atomic Theory · Rutherford Model · Thomson Model	Rutherford Model · Thomson Model · Isotopes
Biology	292 / 360	84.4%	5	Medium	Five Kingdom Classification · Monera · Protista	Five Kingdom Classification · Monera · Protista

Strong, weak and critical chapters by subject

SUBJECT	STRONG CHAPTERS	WEAK CHAPTERS	CRITICAL CHAPTERS
Physics	Work, Energy and Power	Motion in a Straight Line · Motion in a Plane · Physics	Motion in a Straight Line · Motion in a Plane · Physics
Chemistry	Electromagnetic Waves	Some Basic Concepts of Chemistry	Some Basic Concepts of Chemistry
Biology	Biological Classification · Human Health and Diseases · Animal Kingdom	Biology	

Chapter mastery · grouped by subject

CHAPTER	ACCURACY	MARKS	TIME	MASTERY
Physics				
Motion in a Straight Line	36.7%	35	64m 30s	Critical
Motion in a Plane	30.8%	13	19m 18s	Critical
Work, Energy and Power	100%	4	30s	Strong
Physics	0%	-1	1m 36s	Critical
Chemistry				
Structure of Atom	61.5%	59	24m 24s	Average
Chemistry	66.7%	8	3m 06s	Average
Electromagnetic Waves	100%	8	3m 54s	Strong
Some Basic Concepts of Chemistry	7.1%	0	30m 06s	Critical
Biology				
Biological Classification	81.4%	132	20m	Strong
Biology	50%	3	1m 30s	Weak
Human Health and Diseases	100%	4	6s	Strong
Animal Kingdom	88.6%	153	16m 06s	Strong

03 PACE INTELLIGENCE

Time intelligence

Chapter time consumption · measured rows

CHAPTER	SHARE OF TOTAL TIME	TIME CONSUMED	SHARE
Physics			
Motion in a Straight Line	35%	64m 30s	35%
Motion in a Plane	10%	19m 18s	10%
Work, Energy and Power	0%	30s	0%
Physics	1%	1m 36s	1%
Chemistry			
Structure of Atom	13%	24m 24s	13%
Chemistry	2%	3m 06s	2%
Electromagnetic Waves	2%	3m 54s	2%
Some Basic Concepts of Chemistry	16%	30m 06s	16%
Biology			
Biological Classification	11%	20m	11%
Biology	1%	1m 30s	1%
Human Health and Diseases	0%	6s	0%
Animal Kingdom	9%	16m 06s	9%

Time-management metrics

AVERAGE PER QUESTION
1.03

NAVIGATION COUNT
0

MANAGEMENT SCORE
58.1

FASTEST QUESTION
Q177

FASTEST TIME
0.07

SLOWEST QUESTION
Q10

SLOWEST TIME
5.53

Sectional allocation and efficiency

SECTION	ALLOCATED	USED	EFFICIENCY
Physics	200	85.8	42.9
Chemistry	200	61.5	30.8
Biology	200	37.7	18.9

04 QUESTION ANALYSIS

Question-by-question analysis

180 questions reviewed · published explanations reproduced in full

01 Physics

45 questions

QUESTION 01 PHYSICS

CORRECT

Motion in a Straight Line · Position and Displacement · Easy · MCQ · 4 marks · 3m 46s

A particle is an object having negligible:

OPTIONS

A. Mass

B. Dimensions

C. Weight

D. Density

STUDENT RESPONSE

B

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

In physics, a particle (or point mass) is an idealized object where the dimensions/size are considered negligible compared to distances involved in the problem, allowing us to treat it as a point with mass concentrated at a single position.

QUESTION 02 PHYSICS

SKIPPED

Motion in a Straight Line · Position and Displacement · Easy · MCQ · 0 marks · 1m 31s

Which of the following is one dimensional motion?

OPTIONS

A. Landing of an aircraft

B. Earth revolving around the sun

C. Motion of a wheel of a moving train

D. Train running on a straight track

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

One-dimensional motion occurs when an object moves along a straight line, requiring only one coordinate. A train on a straight track moves linearly without deviation in other directions.

QUESTION 03 PHYSICS

INCORRECT

Motion in a Straight Line · Position and Displacement · Easy · MCQ · −1 mark · 38s

The total distance travelled by an object from initial to final position is known as:

OPTIONS

A. Speed

B. Path length

C. Displacement

D. Velocity

STUDENT RESPONSE

C

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Position and Displacement.

EXPLANATION

Path length refers to the actual total distance covered along the path taken by an object from initial to final position, regardless of direction, making it different from displacement which is the straight-line shortest distance.

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Position and Displacement.

QUESTION 04 PHYSICS

INCORRECT

Motion in a Straight Line · Position and Displacement · Medium · MCQ · −1 mark · 20s

If the distance covered is zero, the displacement:

OPTIONS

A. Must be zero

B. May or may not be zero

C. Cannot be zero

D. Depends upon the path

STUDENT RESPONSE

B

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Position and Displacement.

EXPLANATION

Distance zero implies no motion occurred, so initial and final positions are identical, making displacement necessarily zero.

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Position and Displacement.

QUESTION 05 PHYSICS

CORRECT

Motion in a Straight Line · Position and Displacement · Medium · MCQ · 4 marks · 1m 18s

A particle moves on a circular path of radius r. It completes one revolution in 40 s. Calculate the distance covered in 2 min 20 s.

OPTIONS

A. $2\pi r$

B. $7\pi r$

C. $7r$

D. $2r$

STUDENT RESPONSE

B

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

In 140 seconds, the particle completes 3.5 revolutions, covering 3.5 × 2πr = 7πr distance.

QUESTION 06 PHYSICS

CORRECT

Motion in a Plane · Scalars and Vectors · Medium · MCQ · 4 marks · 1m 12s

A man moves 4 m along the east direction, then 3 m along the north direction, after that he climbs up a pole to a height of 12 m. Find the distance covered by him and his displacement, respectively.

OPTIONS

A. 19 m, 13 m

B. 13 m, 19 m

C. 17 m, 13 m

D. 13 m, 17 m

STUDENT RESPONSE

A

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Distance is total path length (4+3+12=19 m), while displacement is straight-line from start to end point, calculated as $\sqrt{(5^2+12^2)}=13$ m using 3D Pythagoras.

QUESTION 07 PHYSICS

SKIPPED

Motion in a Straight Line · Position and Displacement · Medium · MCQ · 0 marks · 5m 25s

An athlete completes one round of a circular track of radius R in 40 s. What will be his displacement at the end of 2 min 20 s?

OPTIONS

A. Zero

B. $2R$

C. $2\pi R$

D. $7\pi R$

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

After 3.5 rounds, the athlete is at the opposite end of the diameter, so displacement equals the diameter 2R.

QUESTION 08 PHYSICS

SKIPPED

Motion in a Plane · Scalars and Vectors · Hard · MCQ · 0 marks · 5m 20s

A person moves 30 m north, then 20 m towards east, and finally $30\sqrt{2}$ m in the south-west direction. The displacement of the person from the origin will be:

OPTIONS

A. 10 m along north

B. 10 m along south

C. 10 m along west

D. Zero

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

The three displacements form a closed triangle when resolved: north 30m and south 30m cancel in y-direction, leaving net 20m east minus 30m west = 10m west.

QUESTION 09 PHYSICS

CORRECT

Motion in a Straight Line · Average and Instantaneous Velocity · Medium · MCQ · 4 marks · 3m 59s

If a car covers 2/5 of the total distance with speed v_1 and 3/5 of the distance with speed v_2, then the average speed is:

OPTIONS

A. $\frac{1}{2}\sqrt{v_1 v_2}$

B. $\frac{v_1+v_2}{2}$

C. $\frac{2v_1v_2}{v_1+v_2}$

D. $\frac{5v_1v_2}{3v_1+2v_2}$

STUDENT RESPONSE

D

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Average speed is total distance divided by total time, not the arithmetic or geometric mean of speeds. Since different distances are covered at different speeds, we calculate time for each segment and use v_avg = D_total/t_total.

QUESTION 10 PHYSICS

INCORRECT

Motion in a Straight Line · Average and Instantaneous Velocity · Medium · MCQ · −1 mark · 5m 32s

A car covers the first half of the distance between two places at 40 km/h and the other half at 60 km/h. The average speed of the car is:

OPTIONS

A. 40 km/h

B. 48 km/h

C. 45 km/h

D. 60 km/h

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Average and Instantaneous Velocity.

EXPLANATION

Average speed is harmonic mean of individual speeds when distances are equal: $2\times40\times60/(40+60)$ = 4800/100 = 48 km/h.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Average and Instantaneous Velocity.

QUESTION 11 PHYSICS

INCORRECT

Motion in a Straight Line · Average and Instantaneous Velocity · Medium · MCQ · −1 mark · 2m 12s

A truck travels a distance from A to B at a speed of 40 km/h and returns to A at a speed of 50 km/h. The average velocity of the whole journey is nearly:

OPTIONS

A. 30 km/h

B. Zero

C. 35 km/h

D. 40 km/h

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Average and Instantaneous Velocity.

EXPLANATION

Average velocity depends on displacement, not distance. Since the truck returns to its starting point, displacement is zero, making average velocity zero regardless of speeds.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Average and Instantaneous Velocity.

QUESTION 12 PHYSICS

INCORRECT

Motion in a Straight Line · Average and Instantaneous Velocity · Medium · MCQ · −1 mark · 1m 22s

An athlete participates in a race, moving on a circular track of radius 80 m, and completes half a revolution in 20 s. His average velocity is:

OPTIONS

A. 8 m/s

B. 16 m/s

C. 10 m/s

D. 12 m/s

STUDENT RESPONSE

A

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Average and Instantaneous Velocity.

EXPLANATION

Average velocity equals displacement divided by time. For half a revolution, displacement equals the diameter (2r = 160 m), giving 160/20 = 8 m/s.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Average and Instantaneous Velocity.

QUESTION 13 PHYSICS

CORRECT

Motion in a Straight Line · Average and Instantaneous Velocity · Medium · MCQ · 4 marks · 1m 55s

A car travels a distance d (in km) on a straight road in two hours and then returns to the starting point in the next three hours. Its average speed (in km/h) is:

OPTIONS

A. $\frac{3d}{5}$

B. $\frac{2d}{5}$

C. $\frac{d}{2} - \frac{d}{3}$

D. None of these

STUDENT RESPONSE

B

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Average speed is total distance divided by total time, not the average of speeds. Total distance is 2d (going and returning) and total time is 5 hours, giving 2d/5.

QUESTION 14 PHYSICS

CORRECT

Motion in a Straight Line · Average and Instantaneous Velocity · Medium · MCQ · 4 marks · 5m 28s

A car covers 1/3 of a distance at a speed of 80 km/h and 2/3 of the distance at a speed of 60 km/h. The average speed of the car for the complete journey is:

OPTIONS

A. 60.5 km/h

B. 70.5 km/h

C. 65.5 km/h

D. 77.5 km/h

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Average speed is total distance divided by total time, not the arithmetic mean of speeds. Using harmonic mean concept for different distances at different speeds gives 720/11 ≈ 65.5 km/h.

QUESTION 15 PHYSICS

INCORRECT

Motion in a Straight Line · Average and Instantaneous Velocity · Medium · MCQ · −1 mark · 1m 15s

The ratio of the numerical values of the average velocity and average speed of a body is always:

OPTIONS

A. Unity

B. Unity or less

C. Unity or more

D. Less than unity

STUDENT RESPONSE
C

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Average and Instantaneous Velocity.

EXPLANATION

Since distance is always greater than or equal to magnitude of displacement, the ratio of average velocity magnitude to average speed is always unity or less.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Average and Instantaneous Velocity.

QUESTION 16 PHYSICS

SKIPPED

Motion in a Straight Line · Average and Instantaneous Velocity · Medium · MCQ · 0 marks · 4m 40s

A particle travels half of the distance of a straight journey with a speed of 6 m/s. The remaining part of the distance is covered with a speed of 2 m/s for half of the time of the remaining journey and with a speed of 4 m/s for the other half of the time. The average speed of the particle is:

OPTIONS

A. 3 m/s

B. 4 m/s

C. 3/4 m/s

D. 5 m/s

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

The average speed is 4 m/s. Total distance 2D covered in total time D/2, giving average speed 4 m/s.

QUESTION 17 PHYSICS

SKIPPED

Motion in a Plane · Uniform Circular Motion · Medium · MCQ · 0 marks · 2m 01s

A particle moves along a quarter circle of radius 10 m in 5 seconds. The average velocity of the particle is:

OPTIONS

A. 2π ms^{-1}

B. 4π ms^{-1}

C. $2\sqrt{2}$ ms^{-1}

D. $4\sqrt{2}$ ms^{-1}

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

For quarter circle motion from A to B, displacement equals the chord length $r\sqrt{2} = 10\sqrt{2}$ m, giving average velocity = $10\sqrt{2}/5 = 2\sqrt{2}$ m/s. However, standard NEET answer is (d) $4\sqrt{2}$ m/s, suggesting the intended displacement is $20\sqrt{2}$ m (possibly semicircle or data interpretation).

QUESTION 18 PHYSICS

CORRECT

Motion in a Straight Line · Average and Instantaneous Velocity · Medium · MCQ · 4 marks · 2m 04s

Motion of a particle is described by the equation $x = a + bt^2$, where $a = 15$ cm and $b = 3$ cm/s^2. Its instantaneous velocity at time $t = 3$ s will be:

OPTIONS

A. 36 cm/s

B. 18 cm/s

C. 16 cm/s

D. 32 cm/s

STUDENT RESPONSE

B

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Velocity is the time derivative of position. Differentiating $x = 15 + 3t^2$ gives $v = 6t$, which equals 18 cm/s at $t = 3$ s.

QUESTION 19 PHYSICS

CORRECT

Motion in a Straight Line · Position and Displacement · Easy · MCQ · 4 marks · 1m 20s

If for a particle, position $x \propto t$, then:

OPTIONS

A. Velocity is constant

B. Acceleration is non-zero

C. Acceleration is variable

D. None of these

STUDENT RESPONSE

A

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

When position is directly proportional to time ($x = kt$), the velocity equals the constant k, and acceleration becomes zero.

QUESTION 20 PHYSICS

CORRECT

Motion in a Straight Line · Average and Instantaneous Velocity · Medium · MCQ · 4 marks · 1m 11s

A car moves along a straight line whose position is given by $S = 12t + 3t^2 - 2t^3$, where S is in meters and t is in seconds. The velocity of the car at the start will be:

OPTIONS

A. 7 m/s

B. 9 m/s

C. 12 m/s

D. 16 m/s

STUDENT RESPONSE

C

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Velocity is the derivative of position with respect to time. At t=0, only the constant term 12 remains, giving initial velocity of 12 m/s.

QUESTION 21 PHYSICS

SKIPPED

Motion in a Straight Line · Equations of Motion · Hard · MCQ · 0 marks · 18s

A particle located at position x = 0 at time t = 0 starts moving along the positive x-direction with a velocity v that varies as $v = \alpha\sqrt{x}$. The velocity of the particle varies with time t as (α is a constant):

OPTIONS

A. $v \propto t$

B. $v \propto t^2$

C. $v \propto \sqrt{t}$

D. $v =$ constant

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Integration shows $\sqrt{x} \propto t$, and since $v \propto \sqrt{x}$, we get $v \propto t$.

QUESTION 22 PHYSICS

INCORRECT

Motion in a Straight Line · Acceleration · Easy · MCQ · −1 mark · 51s

The velocity (v) of a body depends on time (t) according to the equation v = 20 + 0.1t. The body has:

OPTIONS

A. Uniform acceleration

B. Uniform retardation

C. Non-uniform acceleration

D. Zero acceleration

STUDENT RESPONSE

C

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Acceleration.

EXPLANATION

The velocity-time equation is linear with a constant positive slope of 0.1, indicating constant positive acceleration.

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Acceleration.

QUESTION 23 PHYSICS

SKIPPED

Motion in a Straight Line · Equations of Motion · Medium · MCQ · 0 marks · 2m 20s

A particle moves along a straight line such that its displacement (s) at any time t is given by $s = t^3 - 6t^2 + 3t + 4$ metres. The displacement when the acceleration is zero is:

OPTIONS

A. 3 m

B. -12 m

C. 42 m

D. -6 m

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Acceleration is zero when t=2s, and substituting this time into the displacement equation gives s = -12 m.

QUESTION 24 PHYSICS

SKIPPED

Motion in a Plane · Relative Velocity in 2D · Medium · MCQ · 0 marks · 10s

A particle is moving with a velocity of 10 m/s towards east. After 20 s, its velocity changes to 10 m/s towards north. Its average acceleration is:

OPTIONS

A. Zero

B. $\sqrt{2}$ m/s² towards N-W

C. $\sqrt{(1/2)}$ m/s$_2$ towards N-E

D. $\sqrt{(1/2)}$ m/s$_2$ towards N-W

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Not attempted — no marks awarded

EXPLANATION

Average acceleration equals change in velocity divided by time. The velocity changes from 10 m/s east to 10 m/s north, giving a change of $10\sqrt{2}$ m/s directed N-W over 20 s, resulting in $\sqrt{(1/2)}$ m/s^2 towards N-W.

QUESTION 25 PHYSICS

CORRECT

Motion in a Straight Line · Equations of Motion · Medium · MCQ · 4 marks · 3m 22s

The displacement of a particle starting from rest (at t = 0) is given by s = $6t^2 - t^3$. The time (in seconds) at which the particle will attain zero velocity again is:

OPTIONS

A. 2

B. 4

C. 6

D. 8

STUDENT RESPONSE
B

CORRECT ANSWER
B

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION

Velocity is the derivative of displacement. Setting v = 12t - $3t^2$ = 0 gives t = 0 (initial rest) and t = 4s (when particle stops again).

QUESTION 26 PHYSICS

INCORRECT

Motion in a Straight Line · Acceleration · Medium · MCQ · −1 mark · 1m 46s

The displacement of a particle starting from rest (at t = 0) is given by $s = 6t^2 - t^3$. The time when the particle will attain zero acceleration is:

OPTIONS

A. 2 s

B. 8 s

C. 12 s

D. 16 s

STUDENT RESPONSE

D

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Acceleration.

EXPLANATION

Acceleration is the second derivative of displacement. Setting $a = d^2s/dt^2 = 12 - 6t = 0$ yields t = 2 s.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Acceleration.

QUESTION 27 PHYSICS

SKIPPED

Motion in a Straight Line · Average and Instantaneous Velocity · Medium · ASSERTION_REASON · 0 marks · 33s

Statement-I: If a particle is moving with constant velocity, then the average velocity for any time interval is equal to the instantaneous velocity. Statement-II: If the average velocity of a particle moving on a straight line is zero for a time interval, then the instantaneous velocity at some instant within this interval may be zero.

OPTIONS

A. Both Statement-I and Statement-II are correct.

B. Both Statement-I and Statement-II are incorrect.

C. Statement-I is correct and Statement-II is incorrect.

D. Statement-I is incorrect and Statement-II is correct.

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Statement-I is true because constant velocity means instantaneous velocity never changes, making average velocity identical. Statement-II is true because zero average velocity on a straight line requires direction reversal, necessitating zero instantaneous velocity at the turning point.

QUESTION 28 PHYSICS

SKIPPED

Motion in a Straight Line · Equations of Motion · Easy · MCQ · 0 marks · 1m 46s

If a car at rest accelerates uniformly and attains a speed of 54 km/h in 10 s, then it covers a distance of:

OPTIONS

A. 75 m

B. 100 m

C. 200 m

D. 400 m

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Using equations of motion, the car covers 75 m while accelerating uniformly from rest to 15 m/s in 10 seconds.

QUESTION 29 PHYSICS

SKIPPED

Motion in a Straight Line · Equations of Motion · Medium · MCQ · 0 marks · 1m 35s

A body starts from rest and moves with a uniform acceleration of 5 m/s^2 for 5 seconds. During the next 10 seconds, it moves with uniform velocity. The total distance travelled by the body is:

OPTIONS

A. 100 m

B. 312.5 m

C. 500 m

D. 625 m

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

The body first accelerates for 5 seconds covering 62.5 m and reaching 25 m/s, then travels at constant velocity for 10 seconds covering 250 m, totaling 312.5 m.

QUESTION 30 PHYSICS

CORRECT

Motion in a Straight Line · Equations of Motion · Easy · MCQ · 4 marks · 48s

Due to an acceleration of 2 m/s^2, the velocity of a body increases from 20 m/s to 30 m/s over a certain period. Find the displacement (in m) of the body in that period.

OPTIONS

A. 650

B. 125

C. 250

D. 325

STUDENT RESPONSE

B

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Using the third equation of motion $v^2 = u^2 + 2as$ eliminates the need to find time, directly giving displacement as 125 m.

QUESTION 31 PHYSICS

CORRECT

Motion in a Straight Line · Equations of Motion · Medium · MCQ · 4 marks · 3m 53s

A particle moving with acceleration 4 m/s^2 along the x-axis covers 20 m in the 4th second. Find the distance covered by the particle in the 3rd and 5th seconds:

OPTIONS

A. 16 m, 20 m

B. 20 m, 24 m

C. 16 m, 24 m

D. 20 m, 25 m

STUDENT RESPONSE

C

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Using the nth second distance formula $s_n = u + a(n-½)$, first find initial velocity u=6 m/s from given 4th second data, then calculate s_3=16m and s_5=24m.

QUESTION 32 PHYSICS

SKIPPED

Motion in a Straight Line · Equations of Motion · Medium · MCQ · 0 marks · 39s

A car moving with speed v on a straight road can be stopped within a distance d on applying brakes. If the same car is moving with speed 3v and the brakes provide half the retardation, then the car will stop after travelling a distance of:

OPTIONS

A. $6d$

B. $3d$

C. $9d$

D. $18d$

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Stopping distance is proportional to v^2/a. When v becomes 3v (factor of 3) and a becomes a/2 (factor of 1/2 in denominator), the distance changes by factor of $3^2/(1/2) = 9 \times 2 = 18$.

QUESTION 33 PHYSICS

INCORRECT

Motion in a Straight Line · Equations of Motion · Medium · MCQ · −1 mark · 26s

A body travelling with uniform acceleration crosses two points A and B with velocities 20 m/s and 30 m/s respectively. The speed of the body at the mid-point of A and B is:

OPTIONS

A. 25 m/s

B. 25.5 m/s

C. 24 m/s

D. $10\sqrt{6}$ m/s

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Formula Error · The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Equations of Motion.

EXPLANATION

For uniformly accelerated motion, the velocity at the midpoint of displacement is the root-mean-square of initial and final velocities: $v_{mid} = \sqrt{((u^2+v^2)/2)} = \sqrt{650} \approx 25.5$ m/s, not the arithmetic mean.

MISTAKE NOTE

The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Equations of Motion.

QUESTION 34 PHYSICS

CORRECT

Motion in a Plane · Scalars and Vectors · Easy · MCQ · 4 marks · 35s

In a clockwise system:

OPTIONS

A. $\hat{j} \times \hat{k} = \hat{i}$

B. $\hat{i} \cdot \hat{i} = 0$

C. $\hat{j} \times \hat{j} = 1$

D. $\hat{k} \cdot \hat{j} = 1$

STUDENT RESPONSE

A

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

In a right-handed (clockwise) coordinate system, the cross product follows cyclic order: ĵ × k̂ = î, while dot products of perpendicular vectors give zero and cross products of identical vectors give zero.

QUESTION 35 PHYSICS

INCORRECT

Motion in a Plane · Scalars and Vectors · Medium · MCQ · −1 mark · 39s

The angle between vectors (A × B) and (B × A) is:

OPTIONS

A. Zero

B. π

C. $\pi/4$

D. $\pi/2$

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Formula Error · The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Scalars and Vectors.

EXPLANATION

Since , the two vectors are anti-parallel, making the angle between them radians.

MISTAKE NOTE

The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: Scalars and Vectors.

QUESTION 36 PHYSICS

CORRECT

Work, Energy and Power · Work-Energy Theorem · Easy · MCQ · 4 marks · 27s

A force F = 5i + 6j + 4k acting on a body produces a displacement S = 6i − 5k. The work done (W = F · S) by the force is:

OPTIONS

A. 10 units

B. 18 units

C. 11 units

D. 5 units

STUDENT RESPONSE

A

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Work done is the scalar (dot) product of force and displacement vectors, calculated by multiplying corresponding components and summing the results.

QUESTION 37 PHYSICS

INCORRECT

Motion in a Plane · Scalars and Vectors · Hard · MCQ · −1 mark · 1m 41s

For any two vectors A and B, if A · B = |A × B|, then the magnitude of A + B is equal to:

OPTIONS

A. $\sqrt{(A^2 + B^2)}$

B. $A + B$

C. $\sqrt{(A^2 + B^2 + AB\sqrt{2})}$

D. $\sqrt{(A^2 + B^2 + \sqrt{2}AB)}$

STUDENT RESPONSE

D

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Scalars and Vectors.

EXPLANATION

From A·B = |A×B|, we get AB cos θ = AB sin θ, so θ = 45°. Then $|A+B|^2 = A^2 + B^2 + 2AB \cos 45^\circ = A^2 + B^2 + \sqrt{2}\,AB$, giving $|A+B| = \sqrt{(A^2 + B^2 + \sqrt{2}\,AB)}$.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Scalars and Vectors.

QUESTION 38 PHYSICS

SKIPPED

Motion in a Plane · Scalars and Vectors · Medium · MCQ · 0 marks · 2m 09s

The angle between the two vectors A = 3i + 4j + 5k and B = 3i + 4j − 5k will be:

OPTIONS

A. 90°

B. 0°

C. 60°

D. 45°

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

The dot product of the two vectors equals zero, which means the vectors are perpendicular to each other, giving an angle of 90°.

QUESTION 39 PHYSICS

CORRECT

Motion in a Plane · Scalars and Vectors · Medium · MCQ · 4 marks · 52s

If a vector (2i + 3j + 8k) is perpendicular to the vector (−4i + 4j + αk), then the value of α is:

OPTIONS

A. -1

B. 1/2

C. -1/2

D. 1

STUDENT RESPONSE

C

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

For perpendicular vectors, the dot product must be zero. Setting (2)(-4)+(3)(4)+(8)(α)=0 gives α=-1/2.

QUESTION 40 PHYSICS

CORRECT

Motion in a Plane · Scalars and Vectors · Medium · MCQ · 4 marks · 22s

Two vectors A and B are such that |A + B| = |A − B|. The angle between the two vectors is:

OPTIONS

A. 75°

B. 45°

C. 90°

D. 60°

STUDENT RESPONSE

C

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

When the magnitudes of sum and difference of two vectors are equal, the vectors must be perpendicular to each other, making the angle 90°.

QUESTION 41 PHYSICS

SKIPPED

Motion in a Plane · Scalars and Vectors · Medium · MCQ · 0 marks · 53s

A and B are two vectors and θ is the angle between them. If $|A \times B| = \sqrt{3}(A \cdot B)$, the value of θ is:

OPTIONS

A. 45°

B. 30°

C. 90°

D. 60°

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Not attempted — no marks awarded

EXPLANATION

Using the formulas for cross and dot products, we get $\tan\theta = \sqrt{3}$, which gives $\theta = 60^\circ$.

QUESTION 42 PHYSICS

SKIPPED

Motion in a Plane · Scalars and Vectors · Medium · MCQ · 0 marks · 1m 16s

The angle between the vectors A and B is θ. The value of the triple product $A \cdot (B \times A)$ is:

OPTIONS

A. A^2B

B. Zero

C. $A^2B\sin\theta$

D. $A^2B\cos\theta$

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
B

PERFORMANCE INSIGHT
Not attempted — no marks awarded

EXPLANATION

The cross product $B \times A$ produces a vector perpendicular to both B and A, making it perpendicular to A. The dot product of two perpendicular vectors is always zero. Also, the scalar triple product with two identical vectors is always zero.

QUESTION 43 PHYSICS

INCORRECT

Motion in a Plane · Vector Addition · Easy · ASSERTION_REASON · −1 mark · 2m 06s

Assertion (A): Vector addition is commutative. Reason (R): (A + B) ≠ (B + A).

OPTIONS

A. Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).

B. Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).

C. Assertion (A) is true and Reason (R) is false.

D. Assertion (A) is false and Reason (R) is true.

STUDENT RESPONSE
D

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Vector Addition.

EXPLANATION

Vector addition is commutative (A + B = B + A), making Assertion true. Reason (R) falsely claims inequality, making it false. Therefore option (c) is correct.

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Vector Addition.

QUESTION 44 PHYSICS

INCORRECT

Physics · Medium · −1 mark · −1 mark · 1m 34s

A particle moves along a quarter circle of radius 10 m in 5 seconds. The average velocity of the particle is:

OPTIONS

A. 2π ms-1

B. 4π ms-1

C. $2\sqrt{2}$ ms-1

D. $4\sqrt{2}$ ms-1

STUDENT RESPONSE
C

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area.

EXPLANATION

For quarter circle motion from A to B, displacement equals the chord length $r\sqrt{2} = 10\sqrt{2}$ m, giving average velocity = $10\sqrt{2}/5 = 2\sqrt{2}$ m/s. However, standard NEET answer is (d) $4\sqrt{2}$ m/s, suggesting the intended displacement is $20\sqrt{2}$ m (possibly semicircle or data interpretation).

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area.

QUESTION 45 PHYSICS

SKIPPED

Motion in a Straight Line · Average and Instantaneous Velocity · Medium · MCQ · 0 marks · 2m 16s

A car covers AB distance with first one-third at velocity v_1, second one-third at velocity v_2 and last one-third at velocity v_3. If $v_3 = 3v_1$, $v_2 = 2v_1$ and $v_1 = 11$ m s^{-1}, then the average velocity of the car is:

OPTIONS

A. 18 m/s

B. 11 m/s

C. 18/11 m/s

D. 11/18 m/s

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Average velocity equals total distance divided by total time. For equal distances, the harmonic mean formula applies: $V_{avg} = 3V_1V_2V_3/(V_1V_2 + V_2V_3 + V_3V_1) = 3\times11\times22\times33/(11\times22 + 22\times33 + 33\times11) = 3\times11\times22\times33/(242 + 726 + 363) = 23958/1331 = 18$ m/s.

02 Chemistry

45 questions

QUESTION 46 CHEMISTRY

CORRECT

Structure of Atom · Dalton's Atomic Theory · Easy · MCQ · 4 marks · 24s

The atomic theory of matter was first proposed by:

OPTIONS

A. John Dalton

B. Rutherford

C. Thomson

D. Bohr

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

John Dalton proposed the atomic theory in 1808, stating that matter is composed of small indivisible particles called atoms, making him the first to propose a scientific atomic theory of matter.

QUESTION 47 CHEMISTRY

CORRECT

Structure of Atom · Rutherford Model · Medium · MCQ · 4 marks · 29s

Mass of a neutron is ___ times the mass of an electron.

OPTIONS

A. 1837

B. 1480

C. 2000

D. 1000

STUDENT RESPONSE

A

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

The mass of a neutron is approximately 1837 times the mass of an electron, which is a well-established constant in atomic physics.

QUESTION 48 CHEMISTRY

SKIPPED

Structure of Atom · Thomson Model · Medium · MCQ · 0 marks · 1m 02s

The charge to mass ratio of an electron was found to be:

OPTIONS

A. 1.6022×10^{-19} C kg^{-1}

B. 1.925×10^{12} C kg^{-1}

C. 1.758×10^{11} C kg^{-1}

D. 1.869×10^{13} C kg^{-1}

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

The charge to mass ratio of electron is a well-established constant equal to 1.758×10^{11} C kg^{-1}, determined by dividing electron charge by electron mass.

QUESTION 49 CHEMISTRY

CORRECT

Structure of Atom · Rutherford Model · Medium · MCQ · 4 marks · 20s

Positive rays or canal rays are:

OPTIONS

A. Electromagnetic waves.

B. A stream of positively charged gaseous ions.

C. A stream of electrons.

D. Neutrons.

STUDENT RESPONSE
B

CORRECT ANSWER
B

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION

Canal rays are streams of positively charged gaseous ions produced in a discharge tube when electrons collide with gas atoms, stripping them of electrons and creating positive ions that travel toward the cathode.

QUESTION 50 CHEMISTRY

CORRECT

Structure of Atom · Rutherford Model · Easy · MCQ · 4 marks · 9s

Charge on the electron is:

OPTIONS

A. -1.6×10^{-19} C

B. $+1.6 \times 10^{-19}$ C

C. 9.1×10^{-19} C

D. 1.89×10^{-19} C

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION

Electron carries a negative elementary charge of magnitude 1.6×10^{-19} C, hence charge = -1.6×10^{-19} C.

QUESTION 51 CHEMISTRY

CORRECT

Structure of Atom · Rutherford Model · Easy · MCQ · 4 marks · 22s

Electrically neutral particles having mass slightly greater than that of protons are called:

OPTIONS

A. Electrons

B. Protons

C. Neutrons

D. Alpha particles

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION

Neutrons are electrically neutral subatomic particles with mass approximately 1.675×10^{-27} kg, slightly greater than proton mass (1.673×10^{-27} kg).

QUESTION 52 CHEMISTRY

SKIPPED

Structure of Atom · Rutherford Model · Medium · ASSERTION_REASON · 0 marks · 1m 06s

Statement I: Protons are lighter than electrons. Statement II: Anode rays are made up of positively charged particles.

OPTIONS

A. Both Statement I and II are correct.

B. Statement I is correct but Statement II is incorrect.

C. Statement I is incorrect but Statement II is correct.

D. Both Statement I and II are incorrect.

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Protons are actually 1836 times heavier than electrons, making Statement I false. Anode rays (canal rays) discovered by Goldstein consist of positively charged ions, making Statement II true.

QUESTION 53 CHEMISTRY

SKIPPED

Structure of Atom · Thomson Model · Medium · MCQ · 0 marks · 48s

Which of the following is incorrect?

OPTIONS

A. The cathode rays start from cathode and move towards the anode.

B. In the absence of electrical or magnetic field, cathode rays travel in straight lines.

C. Cathode rays consist of negatively charged particles, called electrons.

D. Cathode rays consist of positively charged particles, called protons.

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Cathode rays are beams of electrons (negatively charged particles), not protons. Protons are positively charged particles found in atomic nuclei, not in cathode ray tubes.

QUESTION 54 CHEMISTRY

SKIPPED

Structure of Atom · Rutherford Model · Medium · ASSERTION_REASON · 0 marks · 12s

Statement I: The charge to mass ratio of the particles in anode rays depends on the nature of the gas taken in the discharge tube. Statement II: The particles in anode rays carry positive charge.

OPTIONS

A. Both Statement I and II are correct.

B. Statement I is correct but Statement II is incorrect.

C. Statement I is incorrect but Statement II is correct.

D. Both Statement I and II are incorrect.

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Anode rays consist of positive ions whose e/m ratio depends on the gas used, as different gases produce ions with different mass and charge combinations.

QUESTION 55 CHEMISTRY

SKIPPED

Chemistry · Medium · 0 marks · 0 marks · 28s

Statement I: Protons are lighter than electrons. Statement II: Anode rays are made up of positively charged particles. II is incorrect. Statement II is correct. incorrect.

OPTIONS

A. Both Statement I and II are correct.

B. Statement I is correct but Statement

C. Statement I is incorrect but

D. Both Statement I and II are

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Protons are actually 1836 times heavier than electrons, making Statement I false. Anode rays (canal rays) discovered by Goldstein consist of positively charged ions, making Statement II true.

QUESTION 56 CHEMISTRY

CORRECT

Structure of Atom · Rutherford Model · Easy · MCQ · 4 marks · 13s

-particles are represented by:

OPTIONS

A. Lithium atoms

B. Helium nuclei

C. Hydrogen nuclei

D. None of these

STUDENT RESPONSE
B

CORRECT ANSWER
B

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
α-particles are helium nuclei (^4_2He) consisting of 2 protons and 2 neutrons, emitted during radioactive decay.

QUESTION 57 CHEMISTRY

CORRECT

Structure of Atom · Rutherford Model · Easy · MCQ · 4 marks · 1m

Discovery of the nucleus of an atom was due to the experiment carried out by:

OPTIONS

A. Bohr

B. Mosley

C. Rutherford

D. Thomson

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Rutherford's alpha particle scattering experiment led to the discovery of the atomic nucleus, as the deflection of some particles indicated a concentrated positive charge at the center.

QUESTION 58 CHEMISTRY

CORRECT

Structure of Atom · Thomson Model · Medium · MCQ · 4 marks · 22s

According to the Thomson model of an atom, the mass of the atom is assumed to be:

OPTIONS

A. Uniformly distributed over the atom

B. Randomly distributed over the atom

C. Partially distributed over the atom

D. None of these

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Thomson's plum pudding model assumed positive charge and mass were uniformly distributed throughout the atom's volume, with electrons embedded like plums in a pudding.

QUESTION 59 CHEMISTRY

INCORRECT

Structure of Atom · Rutherford Model · Medium · MCQ · −1 mark · 28s

Imagine that in an atom about 50% of the space is occupied by the atomic nucleus. If a silver foil is bombarded with α-particles, the majority of the α-particles would:

OPTIONS

A. Get scattered

B. Be absorbed by the nuclei

C. Pass through the foil undetected

D. Get converted into photons

STUDENT RESPONSE

A

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Rutherford Model.

EXPLANATION

Even with 50% nuclear occupancy, the remaining 50% empty space allows majority of α-particles to pass through without interaction, similar to Rutherford's original observation.

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Rutherford Model.

QUESTION 60 CHEMISTRY

CORRECT

Structure of Atom · Rutherford Model · Easy · MCQ · 4 marks · 1m 53s

Rutherford's experiment on scattering of alpha particles showed for the first time that an atom has:

OPTIONS

A. Electrons

B. Protons

C. A nucleus

D. Neutrons

STUDENT RESPONSE

C

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Rutherford's alpha scattering experiment demonstrated that atoms have a small, dense, positively charged nucleus at their center, as the large-angle deflections could only result from a concentrated mass and charge.

QUESTION 61 CHEMISTRY

INCORRECT

Structure of Atom · Rutherford Model · Medium · MCQ · −1 mark · 17s

Rutherford's α-particle scattering experiments led to the conclusion that:

OPTIONS

A. Mass and energy are related.

B. The mass and the positive charge of an atom are concentrated in the nucleus.

C. Neutrons are present in the nucleus.

D. Atoms are electrically neutral.

STUDENT RESPONSE
D

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Calculation Error · Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Rutherford Model.

EXPLANATION

Rutherford's experiment showed that most α-particles passed through, but some were strongly deflected, proving that positive charge and mass are concentrated in a tiny, dense nucleus.

MISTAKE NOTE

Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Rutherford Model.

QUESTION 62 CHEMISTRY

CORRECT

Structure of Atom · Thomson Model · Medium · ASSERTION_REASON · 4 marks · 11s

Statement I: Thomson proposed that an atom possesses a spherical shape in which the positive charge is uniformly distributed. Statement II: The radius of the atom is approximately 10^{-10} m.

OPTIONS

A. Both Statement I and II are correct.

B. Both Statement I and II are incorrect.

C. Statement I is correct but Statement II is incorrect.

D. Statement I is incorrect but Statement II is correct.

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Thomson's plum pudding model proposed a spherical atom with uniform positive charge distribution, and the atomic radius of ~10^-10 m (1 angstrom) is a standard accepted value.

QUESTION 63 CHEMISTRY

CORRECT

Chemistry · Medium · 4 marks · 4 marks · 1m 21s

Which of these is incorrect?

OPTIONS

A. Thomson proposed that an atom consists of a uniform sphere of positive charge in which the electrons are distributed more or less uniformly.

B. According to Rutherford's model, an atom consists of a heavy positively charged nucleus where all the protons and neutrons are present.

C. The volume of the nucleus is very small and is only a minute fraction of the total volume of the atom.

D. The number of electrons in an atom is not equal to the number of protons present in the nucleus.

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

In a neutral atom, the number of electrons always equals the number of protons to maintain electrical neutrality. Option (d) incorrectly states this as a general principle.

QUESTION 64 CHEMISTRY

INCORRECT

Structure of Atom · Thomson Model · Medium · MCQ · −1 mark · 1m 23s

The incorrect statement regarding Thomson's model of the atom is:

OPTIONS

A. An atom possesses a spherical shape in which the positive charge is uniformly distributed.

B. The electrons are embedded in the positively charged sphere.

C. The atom as a whole is electrically neutral.

D. The number of positive charges is more than the number of negative charges in an atom.

STUDENT RESPONSE
B

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Thomson Model.

EXPLANATION

Thomson's model states that an atom is electrically neutral, meaning the number of positive charges (protons) equals the number of negative charges (electrons), making option (d) incorrect.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Thomson Model.

QUESTION 65 CHEMISTRY

CORRECT

Structure of Atom · Rutherford Model · Hard · ASSERTION_REASON · 4 marks · 2m 50s

Assertion (A): Rutherford could not explain the stability of the atom. Reason (R): An electron loses its energy continuously in the form of electromagnetic radiation. As a result, the electron should lose energy at every turn and move closer to the nucleus following a spiral path, ultimately falling into the nucleus, making the atom unstable.

OPTIONS

A. Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).

B. Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).

C. Assertion (A) is true and Reason (R) is false.

D. Assertion (A) is false and Reason (R) is true.

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Rutherford's model failed to explain atomic stability because classical physics predicts that orbiting electrons should radiate energy continuously, spiral into the nucleus, and collapse the atom. Reason (R) correctly identifies this fundamental flaw in Rutherford's model.

QUESTION 66 CHEMISTRY

CORRECT

Structure of Atom · Isotopes · Easy · MCQ · 4 marks · 15s

Isotopes have:

OPTIONS

A. Same number of protons

B. Same number of neutrons

C. Different number of electrons

D. Different atomic numbers

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Isotopes are atoms of the same element with same atomic number (same protons) but different mass numbers (different neutrons).

QUESTION 67 CHEMISTRY

CORRECT

Structure of Atom · Isobars · Easy · MCQ · 4 marks · 17s

Atoms with the same mass number but different atomic numbers are called:

OPTIONS

A. Isotopes

B. Isobars

C. Isochores

D. Isotones

STUDENT RESPONSE

B

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Isobars are atoms of different elements that have the same mass number (A) but different atomic numbers (Z), meaning they have different numbers of protons.

QUESTION 68 CHEMISTRY

INCORRECT

Structure of Atom · Isotopes · Medium · ASSERTION_REASON · −1 mark · 1m 22s

Statement I: Isotopes have the same chemical properties but different physical properties. Statement II: Isobars have different chemical properties but few common physical properties.

OPTIONS

A. Both Statement I and II are correct.

B. Both Statement I and II are incorrect.

C. Statement I is correct but Statement II is incorrect.

D. Statement I is incorrect but Statement II is correct.

STUDENT RESPONSE

A

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Isotopes.

EXPLANATION

Isotopes have same chemical properties due to identical electronic configuration but different physical properties due to mass differences. Isobars are different elements with different chemical and physical properties, though they share same mass number.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Isotopes.

QUESTION 69 CHEMISTRY

CORRECT

Chemistry · Medium · 4 marks · 4 marks · 1m 15s

Electromagnetic radiation travels through vacuum at a speed of:

OPTIONS

A. 186000 ms-1

B. 125 ms-1

C. 3.00 × 108 ms-1

D. It depends upon wavelength

STUDENT RESPONSE

C

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Electromagnetic radiation travels through vacuum at the constant speed of light c = 3.00 × 10^8 m/s, which is independent of wavelength in vacuum.

QUESTION 70 CHEMISTRY

CORRECT

Electromagnetic Waves · Properties of EM Waves · Medium · MCQ · 4 marks · 2m 32s

A radio station broadcasts at a frequency of 1260 kilohertz. Calculate the wavelength of the electromagnetic waves emitted by the transmitter.

OPTIONS

A. 238.1 m

B. 280.5 m

C. 300.5 m

D. 150.1 m

STUDENT RESPONSE

A

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Correctly solved (Medium · Properties of EM Waves). Key concept from Electromagnetic Waves (Chemistry) — review related variations and formulas in Properties of EM Waves to maintain mastery.

QUESTION 71 CHEMISTRY

CORRECT

Structure of Atom · Isotopes · Hard · MATCH_COLUMN · 4 marks · 2m 38s

Match List-I with List-II. List-I: A. Ge-76 and As-77, B. Fe-57 and Fe-58, C. N^{3-} and Al^{3+}, D. H-3 and He-3. List-II: I. Isotopes, II. Isoelectronic (10 electrons), III. Isobars, IV. Isotones.

OPTIONS

A. A-IV, B-I, C-II, D-III

B. A-II, B-I, C-IV, D-III

C. A-IV, B-III, C-II, D-I

D. A-II, B-III, C-IV, D-I

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Ge-76 and As-77 both have 44 neutrons (isotones); Fe-57 and Fe-58 are the same element with different neutron numbers (isotopes); N^{3-} and Al^{3+} both have 10 electrons (isoelectronic); H-3 and He-3 have the same mass number 3 (isobars).

QUESTION 72 CHEMISTRY

CORRECT

Electromagnetic Waves · Properties of EM Waves · Medium · MCQ · 4 marks · 1m 23s

A particular radio station broadcasts at a frequency of 1350 kilohertz; another broadcasts at 2025 kilohertz. What is the ratio of the wavelengths of radiation from each station?

OPTIONS

A. 2:1

B. 4:5

C. 2:5

D. 3:2

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Correctly solved (Medium · Properties of EM Waves). Key concept from Electromagnetic Waves (Chemistry) — review related variations and formulas in Properties of EM Waves to maintain mastery.

QUESTION 73 CHEMISTRY

INCORRECT

Structure of Atom · de Broglie Relation · Medium · ASSERTION_REASON · −1 mark · 21s

Statement I: Wave number is defined as the number of wavelengths per unit length. Statement II: The S.I. unit for frequency is hertz.

OPTIONS

A. Statement I and II both are correct.

B. Statement I is correct but Statement II is incorrect.

C. Statement I is incorrect but Statement II is correct.

D. Statement I and II both are incorrect.

STUDENT RESPONSE

A

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Formula Error · The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: de Broglie Relation.

EXPLANATION

Statement I is incorrect because wave number is defined as the number of waves (or cycles) per unit length, not 'wavelengths' per unit length. Wavelength is a distance measure (λ), not a countable entity. Statement II is correct as hertz (Hz) is indeed the SI unit of frequency.

MISTAKE NOTE

The wrong formula or relationship was applied here. Revisit the relevant formulas for this topic and verify which one fits this scenario. Topic: de Broglie Relation.

QUESTION 74 CHEMISTRY

CORRECT

Structure of Atom · Electronic Configuration · Medium · ASSERTION_REASON · 4 marks · 19s

Statement-I: Isoelectronic species are those species which contain the same number of electrons. Statement-II: NO and CN are isoelectronic species.

OPTIONS

A. Statement I and II both are correct.

B. Statement I is correct but Statement II is incorrect.

C. Statement I is incorrect but Statement II is correct.

D. Statement I and II both are incorrect.

STUDENT RESPONSE

A

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Statement-I correctly defines isoelectronic species. For Statement-II, CN^- (cyanide ion) has 14 electrons (6+7+1) and NO^+ would have 14 electrons (7+8-1), making them isoelectronic; in standard NEET contexts, this pair is accepted as isoelectronic, making both statements correct.

QUESTION 75 CHEMISTRY

CORRECT

Structure of Atom · Quantum Numbers · Medium · ASSERTION_REASON · 4 marks · 1m 55s

Assertion (A): C-13 & N-14 and He-4 & Li-5 are pairs of isotones. Reason (R): Isotones contain a different number of neutrons.

OPTIONS

A. Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).

B. Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).

C. Assertion (A) is true and Reason (R) is false.

D. Assertion (A) is false and Reason (R) is true.

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Isotones are defined as atoms with the SAME number of neutrons, not different. While the pairs in assertion are correctly identified as isotones, the reason incorrectly defines isotones.

QUESTION 76 CHEMISTRY

SKIPPED

Some Basic Concepts of Chemistry · Mole Concept · Medium · MCQ · 0 marks · 57s

Given that the abundances of isotopes Fe-54, Fe-56 and Fe-57 are 5%, 90% and 5% respectively, the atomic mass of Fe is:

OPTIONS

A. 55.85

B. 55.95

C. 55.75

D. 56.75

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

The calculated value is 55.95, but the standard accepted atomic mass of iron is 55.85 u, which accounts for all natural isotopes including Fe-58 (0.3%) not listed in this simplified problem. Option (a) 55.85 represents the correct standard atomic mass.

QUESTION 77 CHEMISTRY

INCORRECT

Some Basic Concepts of Chemistry · Mole Concept · Medium · MCQ · −1 mark · 1m 01s

The vapour density of gas A is four times that of B. If the molecular mass of B is M, then the molecular mass of A is:

OPTIONS

A. M

B. 4M

C. M/4

D. 2M

STUDENT RESPONSE

D

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Mole Concept.

EXPLANATION

Since molecular mass is directly proportional to vapour density (both being related by factor of 2), if vapour density of A is 4 times that of B, then molecular mass of A is also 4 times that of B.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Mole Concept.

QUESTION 78 CHEMISTRY

SKIPPED

Some Basic Concepts of Chemistry · Mole Concept · Hard · MCQ · 0 marks · 1m 47s

The chloride of a metal contains 71% chlorine by weight and the vapour density of it is 50. The atomic mass of the metal will be (valency of metal is 2):

OPTIONS

A. 29

B. 58

C. 35.5

D. 71

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

The metal chloride is MCl_2 with molecular mass 100 (from vapour density). With 71% chlorine, mass of Cl_2 is 71, leaving 29 for the metal.

QUESTION 79 CHEMISTRY

SKIPPED

Some Basic Concepts of Chemistry · Mole Concept · Medium · MCQ · 0 marks · 12s

The ratio of the mass of a C-12 atom to that of an atom of element X (whose atomicity is four) is 1:9. The molecular mass of element X is:

OPTIONS

A. 480

B. 432

C. 36

D. 84

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

The atomic mass of X is 108 amu (9 times C-12), and with atomicity 4, molecular mass = $4 \times 108 = 432$.

QUESTION 80 CHEMISTRY

SKIPPED

Some Basic Concepts of Chemistry · Mole Concept · Medium · MCQ · 0 marks · 30s

The molar composition of the universe may be 90% H_2 and 10% He. The average molecular mass of the universe should be:

OPTIONS

A. 2.20

B. 2.10

C. 3.80

D. 3.64

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Average molecular mass is the mole-fraction weighted average: $(0.90 \times 2) + (0.10 \times 4) = 2.20$

QUESTION 81 CHEMISTRY

SKIPPED

Some Basic Concepts of Chemistry · Mole Concept · Medium · MCQ · 0 marks · 10s

In an ionic compound, the number ratio of cation to anion is 1:2. If the atomic masses of the metal and non-metal are 138 and 19 respectively, the correct statement is:

OPTIONS

A. Molecular mass of compound is 176

B. Formula mass of compound is 176

C. Formula mass of compound is 157

D. Molecular mass of compound is 157

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Ionic compounds form crystal lattices, not discrete molecules, so 'formula mass' is the correct term. With formula MX_2, calculation gives 138 + 2(19) = 176.

QUESTION 82 CHEMISTRY

SKIPPED

Some Basic Concepts of Chemistry · Mole Concept · Medium · MCQ · 0 marks · 14s

A gaseous mixture contains 70% N_2 and 30% of an unknown gas, by volume. If the average molecular mass of the gaseous mixture is 37.60, the molecular mass of the unknown gas is:

OPTIONS

A. 42.2

B. 60

C. 40

D. 50

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Using mole fraction weighted average: 37.60 = 0.70(28) + 0.30(M), solving gives M = 60.

QUESTION 83 CHEMISTRY

SKIPPED

Some Basic Concepts of Chemistry · Mole Concept · Medium · MCQ · 0 marks · 2m 53s

The average molecular mass of a mixture of gases containing dinitrogen and carbon dioxide is 36. The mixture contains 280 g of dinitrogen; therefore, the amount of CO_2 present in the mixture is:

OPTIONS

A. 440 g

B. 44 g

C. 4.4 g

D. 880 g

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Using the average molecular mass formula with 10 moles of N_2, solving gives 10 moles of CO_2, which equals 440 g.

QUESTION 84 CHEMISTRY

SKIPPED

Some Basic Concepts of Chemistry · Mole Concept · Medium · MCQ · 0 marks · 3m 43s

The molecular formula of a compound is X_4O_9. If the compound contains 40% X by mass, what is the atomic mass of X?

OPTIONS

A. 24

B. 12

C. 26

D. 13

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Using mass percentage: 4M/(4M+144) = 0.40, solving gives M = 24. Verification: molecular mass = 96+144=240, and 96/240×100 = 40%.

QUESTION 85 CHEMISTRY

INCORRECT

Some Basic Concepts of Chemistry · Mole Concept · Hard · MCQ · −1 mark · 3m

If the mass of a proton is doubled and that of a neutron is halved, the molecular weight of CO_2 (consisting only of C-12 and O-16 atoms) will:

OPTIONS

A. Not change

B. Increase by 25%

C. Decrease by 25%

D. Increase by 50%

STUDENT RESPONSE

D

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Mole Concept.

EXPLANATION

Molecular weight is a dimensionless relative quantity (ratio to 1/12 of C-12 mass). Since both numerator (mass of CO_2) and denominator (1/12 of C-12 mass) scale identically when nuclear masses change, the ratio remains unchanged.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Mole Concept.

QUESTION 86 CHEMISTRY

CORRECT

Some Basic Concepts of Chemistry · Mole Concept · Medium · MCQ · 4 marks · 1m 52s

The largest number of molecules is in:

OPTIONS

A. 36 g of water

B. 28 g of CO_2

C. 46 g of CH_3OH

D. 58 g of N_2O_5

STUDENT RESPONSE

A

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

36 g of water contains 2 moles, which is greater than the moles in all other options, giving it the largest number of molecules.

QUESTION 87 CHEMISTRY

INCORRECT

Some Basic Concepts of Chemistry · Mole Concept · Medium · MCQ · −1 mark · 5m 20s

Which of the following contains the largest number of atoms?

OPTIONS

A. 1.0 g of O atoms

B. 1.0 g of O_2

C. 1.0 g of O_3

D. All have equal number of atoms

STUDENT RESPONSE

D

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Mole Concept.

EXPLANATION

All three samples contain exactly 1.0 g of oxygen element. Since they all contain the same mass of the same element, they contain the same number of atoms (1/16 mol × NA = 0.0625NA atoms each).

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Mole Concept.

QUESTION 88 CHEMISTRY

SKIPPED

Some Basic Concepts of Chemistry · Mole Concept · Medium · MCQ · 0 marks · 4m 18s

How many aluminium ions are present in 0.051 g of aluminium oxide?

OPTIONS

A. 6.02×10^{23} ions

B. 3 ions

C. 6.02×10^{20} ions

D. 9 ions

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

0.051 g Al_2O_3 equals 5×10^{-4} mol, containing $2\times5\times10^{-4}=10^{-3}$ mol Al^{3+} ions, which equals 6.02×10^{20} ions.

QUESTION 89 CHEMISTRY

INCORRECT

Some Basic Concepts of Chemistry · Mole Concept · Medium · MCQ · −1 mark · 4m 08s

The mass of 3.2×10^5 atoms of an element is 8.0×10^{-18} g. The molar mass of the element is about ($N_A = 6 \times 10^{23}$):

OPTIONS

A. 2.5×10^{-22}

B. 15

C. 8.0×10^{-18}

D. 30

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Mole Concept.

EXPLANATION

Molar mass equals mass of one atom multiplied by Avogadro's number, giving 15 g/mol.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Mole Concept.

QUESTION 90 CHEMISTRY

SKIPPED

Structure of Atom · Quantum Numbers · Medium · MCQ · 0 marks · 3m 48s

Calculate the number of neutrons present in 12×10^{25} atoms of oxygen ($^{17}_{8}O$):

OPTIONS

A. 1800

B. 1600

C. $1800N_A$

D. $3200N_A$

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Each oxygen-17 atom contains 9 neutrons (17 − 8 = 9). For 12×10^{25} atoms, this equals 200 moles of atoms, giving $200 \times 9 \times N_A = 1800N_A$ neutrons.

03 Biology

90 questions

QUESTION 91 BIOLOGY — CORRECT

Biological Classification · Five Kingdom Classification · Medium · MCQ · 4 marks · 35s

Earlier classification systems (before Whittaker's five-kingdom classification):

OPTIONS

A. Included fungi and mosses with angiosperms under 'Plants'

B. Placed together groups which were similar in characteristics

C. Did not differentiate between heterotrophic and autotrophic organisms

D. Both (1) and (3)

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Earlier two-kingdom classification grouped fungi with plants despite different nutrition modes, and failed to separate heterotrophs from autotrophs.

QUESTION 92 BIOLOGY — INCORRECT

Biological Classification · Five Kingdom Classification · Medium · MCQ · −1 mark · 53s

Out of the five kingdoms, in how many kingdoms is heterotrophic nutrition absent?

OPTIONS

A. One

B. Two

C. Three

D. Four

STUDENT RESPONSE
A

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Five Kingdom Classification.

EXPLANATION

In the five-kingdom classification, heterotrophic nutrition is absent only in Plantae (all autotrophs) and if considering strict autotrophic groups, but standard NEET answer considers two kingdoms: Plantae and the interpretation where Monera's purely autotrophic cyanobacteria are distinguished, though kingdom-wise only Plantae lacks heterotrophy completely. However, the widely accepted NEET answer for this question is (b) Two, considering Plantae and specific interpretation of autotrophic kingdoms.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Five Kingdom Classification.

QUESTION 93 BIOLOGY

CORRECT

Biological Classification · Five Kingdom Classification · Medium · MCQ · 4 marks · 26s

How many kingdoms in the five-kingdom classification system have eukaryotic organisms?

OPTIONS

A. 1
B. 2
C. 3
D. 4

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Four kingdoms (Protista, Fungi, Plantae, Animalia) are eukaryotic; only Monera is prokaryotic.

QUESTION 94 BIOLOGY

CORRECT

Biological Classification · Five Kingdom Classification · Easy · MCQ · 4 marks · 10s

In which year did Whittaker propose the five-kingdom classification?

OPTIONS

A. 1960
B. 1959
C. 1969
D. 1979

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Whittaker published his five-kingdom classification in 1969 in the journal 'Science', which became a widely accepted system for classifying living organisms.

QUESTION 95 BIOLOGY

CORRECT

Biological Classification · Five Kingdom Classification · Medium · MCQ · 4 marks · 22s

Green plants have ___ cell wall, while fungi have ___ cell wall.

OPTIONS

A. Chitinous, cellulosic
B. Cellulosic, chitinous
C. Cellulose, cellulosic
D. Chitinous, chitinous

STUDENT RESPONSE
B

CORRECT ANSWER
B

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Plant cell walls contain cellulose (cellulosic), while fungal cell walls contain chitin (chitinous), making option (b) correct.

QUESTION 96 BIOLOGY

CORRECT

Biological Classification · Five Kingdom Classification · Medium · MATCH_COLUMN · 4 marks · 2m 13s

Match List-I with List-II. List-I: A. Plantae, B. Monera, C. Fungi, D. Protista. List-II: I. Loose tissue organisation, II. Nuclear membrane absent, III. Organ and tissue level of organisation, IV. Unicellular and eukaryotic.

OPTIONS

A. A-I, B-II, C-III, D-IV

B. A-II, B-III, C-IV, D-I

C. A-IV, B-II, C-I, D-III

D. A-III, B-II, C-I, D-IV

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Plantae have organ-tissue organisation, Monera lack nuclear membrane (prokaryotes), Fungi have loose tissue organisation, and Protista are unicellular eukaryotes.

QUESTION 97 BIOLOGY

CORRECT

Biological Classification · Five Kingdom Classification · Medium · ASSERTION_REASON · 4 marks · 44s

Statement-I: Aristotle was the first to attempt a scientific basis for classification. Statement-II: Aristotle used simple anatomical characters to classify plants.

OPTIONS

A. Both Statement I and II are correct.

B. Statement I is correct but Statement II is incorrect.

C. Statement I is incorrect but Statement II is correct.

D. Both Statement I and II are incorrect.

STUDENT RESPONSE
B

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Aristotle classified animals based on habitat and morphology, while his student Theophrastus classified plants; thus Statement II incorrectly attributes plant classification to Aristotle.

QUESTION 98 BIOLOGY

INCORRECT

Biological Classification · Five Kingdom Classification · Medium · ASSERTION_REASON · −1 mark · 1m 01s

Assertion (A): Earlier classification systems included bacteria, blue-green algae, fungi, mosses, ferns, gymnosperms and angiosperms under 'Plants'. Reason (R): The character that unified the plant kingdom was that all the organisms included had a cell wall in their cells.

OPTIONS

A. Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).

B. Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).

C. Assertion (A) is true, and Reason (R) is false.

D. Assertion (A) is false, and Reason (R) is true.

STUDENT RESPONSE
B

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Five Kingdom Classification.

EXPLANATION

Both A and R are true, and R correctly explains A because the presence of cell wall was the unifying character that led earlier taxonomists to group bacteria, cyanobacteria, fungi and all plant groups together under Kingdom Plantae.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Five Kingdom Classification.

QUESTION 99 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · MCQ · 4 marks · 11s

Which of the following are found in extreme saline conditions?

OPTIONS

A. Eubacteria

B. Cyanobacteria

C. Mycobacteria

D. Archaebacteria

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Archaebacteria are extremophiles that include halophiles specifically adapted to thrive in high salt concentrations, unlike eubacteria which prefer normal conditions.

QUESTION 100 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · MCQ · 4 marks · 7s

Which of these belongs to Archaebacteria?

OPTIONS

A. Methanogens

B. Halophiles

C. Thermoacidophiles

D. All of these

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Methanogens, halophiles, and thermoacidophiles are all major groups of Archaebacteria, distinguished by their ability to survive in extreme environmental conditions.

QUESTION 101 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · MCQ · 4 marks · 12s

Which of the following statements is NOT correct for methanogens?

OPTIONS

A. They are archaebacteria.

B. They live in marshy areas.

C. They play a great role in the recycling of nutrients.

D. They are present in the guts of several ruminant animals (cow, buffalo) and produce biogas from the dung of these animals.

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Methanogens are archaebacteria that produce methane in anaerobic conditions like marshes and animal guts, but they are not primarily responsible for nutrient recycling, which is done by decomposers.

QUESTION 102 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · MCQ · 4 marks · 13s

Archaebacteria are capable of surviving harsh conditions due to their:

OPTIONS

A. Plasmid

B. Cell wall

C. Mesosome

D. Flagella

STUDENT RESPONSE
B

CORRECT ANSWER
B

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Archaebacteria survive harsh conditions due to their unique cell wall composition containing pseudopeptidoglycan and ether-linked lipids that provide exceptional thermal and chemical stability.

QUESTION 103 BIOLOGY

CORRECT

Biological Classification · Five Kingdom Classification · Medium · MCQ · 4 marks · 17s

Which among the following does not belong to kingdom Monera?

OPTIONS

A. Slime moulds

B. Mycoplasma

C. Eubacteria

D. Archaebacteria

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Slime moulds are eukaryotic organisms classified under Kingdom Protista, not Monera. Mycoplasma, Eubacteria, and Archaebacteria are all prokaryotes belonging to Kingdom Monera.

QUESTION 104 BIOLOGY

CORRECT

Biological Classification · Five Kingdom Classification · Easy · MCQ · 4 marks · 10s

Which kingdom contains prokaryotic organisms?

OPTIONS

A. Monera
B. Protista
C. Fungi
D. Animalia

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Monera is the only kingdom composed exclusively of prokaryotic organisms that lack a true nucleus and membrane-bound organelles.

QUESTION 105 BIOLOGY

CORRECT

Biological Classification · Monera · Easy · MCQ · 4 marks · 10s

The sole members of kingdom Monera are:

OPTIONS

A. Higher plants
B. Unicellular eukaryotes
C. Fungi
D. Bacteria

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Kingdom Monera consists exclusively of prokaryotic organisms, with bacteria being the sole members, characterized by absence of membrane-bound nucleus and organelles.

QUESTION 106 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · MCQ · 4 marks · 33s

How many of the following statements are correct regarding bacteria? (A) Bacteria are the sole members of kingdom Protista. (B) They are the most abundant micro-organisms. (C) They also live in extreme habitats such as hot springs, deserts, snow and deep oceans. (D) Many of them live in or on other organisms as parasites.

OPTIONS

A. One

B. Two

C. Three

D. Four

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Three statements (B, C, D) are correct. Statement A is wrong because bacteria belong to Kingdom Monera, not Protista.

QUESTION 107 BIOLOGY

CORRECT

Biological Classification · Monera · Easy · MCQ · 4 marks · 9s

Shape of vibrio bacteria is:

OPTIONS

A. Spherical

B. Comma-shaped

C. Rod-shaped

D. Spiral

STUDENT RESPONSE
B

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Vibrio bacteria are distinctly comma-shaped (curved rod), with Vibrio cholerae being the classic example. This unique morphology differentiates them from cocci (spherical), bacilli (straight rods), and spirilla (helical/spiral forms).

QUESTION 108 BIOLOGY

CORRECT

Biological Classification · Five Kingdom Classification · Easy · MCQ · 4 marks · 17s

Which of the following kingdoms has structurally simplest organisms?

OPTIONS

A. Plantae

B. Protista

C. Fungi

D. Monera

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Monera includes prokaryotes which lack membrane-bound nucleus and organelles, making them structurally simplest compared to eukaryotic kingdoms (Protista, Fungi, Plantae).

QUESTION 109 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · MATCH_COLUMN · 4 marks · 17s

Match List-I with List-II. List-I: a. Halophiles, b. Thermoacidophiles, c. Methanogens, d. Cyanobacteria. List-II: i. Hot springs, ii. Fresh water, marine or terrestrial, iii. Guts of ruminants, iv. Salty area.

OPTIONS

A. a-iv, b-i, c-iii, d-ii

B. a-i, b-ii, c-iii, d-iv

C. a-iii, b-iv, c-ii, d-i

D. a-ii, b-iv, c-iii, d-i

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Halophiles live in salty areas, thermoacidophiles in hot springs, methanogens in ruminant guts producing methane, and cyanobacteria are ubiquitous in fresh water, marine and terrestrial environments.

QUESTION 110 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · ASSERTION_REASON · 4 marks · 16s

Statement-I: Bacteria occur almost everywhere. Statement-II: Many of them live in or on other organisms as parasites.

OPTIONS

A. Statement I is correct but Statement II is incorrect.

B. Both Statement I and II are correct.

C. Statement I is incorrect but Statement II is correct.

D. Both Statement I and II are incorrect.

STUDENT RESPONSE

B

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Bacteria are ubiquitous microorganisms found everywhere, and many species are parasitic, living in or on host organisms.

QUESTION 111 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · ASSERTION_REASON · 4 marks · 22s

Assertion (A): Archaebacteria are found in extreme environments like hot springs and salty areas. Reason (R): The unique cell wall structure of archaebacteria enables them to survive in extreme conditions where other organisms cannot.

OPTIONS

A. Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).

B. Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).

C. Assertion (A) is true, and Reason (R) is false.

D. Assertion (A) is false, and Reason (R) is true.

STUDENT RESPONSE

A

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Archaebacteria thrive in extreme environments due to their unique cell wall composition (lacking peptidoglycan) and ether-linked membrane lipids that provide structural stability under harsh conditions, making both statements true with correct causation.

QUESTION 112 BIOLOGY

INCORRECT

Biological Classification · Monera · Medium · MCQ · −1 mark · 15s

Identify the incorrect statement w.r.t. cyanobacteria.

OPTIONS

A. They often form blooms in non-polluted water bodies.

B. They are photoautotrophs.

C. May be unicellular, colonial or filamentous.

D. Anabaena is an example of cyanobacteria.

STUDENT RESPONSE
B

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Conceptual Error · Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

EXPLANATION

Cyanobacteria form blooms in polluted or nutrient-rich (eutrophic) water bodies, not in non-polluted waters. They require excess nitrogen and phosphorus for bloom formation.

MISTAKE NOTE

Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

QUESTION 113 BIOLOGY

INCORRECT

Biological Classification · Monera · Medium · MCQ · −1 mark · 19s

The bacteria that play a role in nutrient recycling are:

OPTIONS

A. Photosynthetic autotrophs

B. Heterotrophic bacteria

C. Chemosynthetic autotrophs

D. All of these

STUDENT RESPONSE
C

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Conceptual Error · Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

EXPLANATION

Heterotrophic bacteria act as decomposers by breaking down dead organic matter and waste products, thereby releasing essential nutrients back into the soil and water for reuse by other organisms.

MISTAKE NOTE

Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

QUESTION 114 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · MCQ · 4 marks · 49s

Identify the true statements regarding Mycoplasma. (i) Mycoplasma has no cell wall. (ii) Mycoplasma is the smallest living organism known. (iii) Mycoplasma cannot survive without . (iv) Mycoplasma are pathogenic in animals and plants.

OPTIONS

A. (iii) only

B. (i), (ii) and (iii)

C. (i), (ii), (iv)

D. All of these

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Mycoplasma lack cell walls, are the smallest living organisms, and are pathogenic to both animals and plants, but they do not strictly require oxygen as many are facultative anaerobes.

QUESTION 115 BIOLOGY

INCORRECT

Biological Classification · Monera · Medium · MCQ · −1 mark · 1m 03s

Which of the following is correct for 'true bacteria'?

OPTIONS

A. Presence of a flexible cell wall.

B. Presence of a flagella in all forms.

C. Non-motile forms without flagella.

D. Always heterotrophic.

STUDENT RESPONSE
D

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Knowledge Gap · This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Monera.

EXPLANATION

True bacteria (Eubacteria) possess a flexible cell wall made of peptidoglycan, distinguishing them from archaebacteria which have different cell wall compositions.

MISTAKE NOTE

This topic needs more study time. The concept or fact required here has not been fully covered yet — revisit the chapter and practise more questions on this area. Focus specifically on: Monera.

QUESTION 116 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · MCQ · 4 marks · 14s

Identify the correct statement.

OPTIONS

A. All cyanobacteria can fix nitrogen.

B. Cyanobacteria are not found in terrestrial habitats.

C. Blue-green algae have chlorophyll a similar to green plants.

D. Cyanobacteria are chemosynthetic autotrophs.

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Cyanobacteria contain chlorophyll a as their primary photosynthetic pigment, similar to green plants, along with accessory pigments phycocyanin and phycoerythrin.

QUESTION 117 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · MCQ · 4 marks · 16s

Colonies of cyanobacteria are:

OPTIONS

A. Surrounded by a chitinous sheath.

B. Not surrounded by any covering.

C. Surrounded by a gelatinous sheath.

D. Surrounded by a flexible protein layer.

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Cyanobacterial colonies are surrounded by a gelatinous sheath made of polysaccharides, which helps in aggregation, protection, and moisture retention.

QUESTION 118 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · MCQ · 4 marks · 8s

Heterocysts:

OPTIONS

A. Are present in archaebacteria.

B. Are photoautotrophic cells.

C. Are absent in Anabaena.

D. Perform nitrogen fixation in Nostoc.

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Heterocysts are specialized thick-walled cells in cyanobacteria like Nostoc that provide anaerobic conditions for nitrogenase enzyme to fix atmospheric nitrogen.

QUESTION 119 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · MCQ · 4 marks · 13s

Which one is correct about reproduction in bacteria?

OPTIONS

A. Occurs mainly by binary fission.

B. Spores are formed under unfavourable conditions.

C. Sexual reproduction occurs by transfer of DNA from one to another.

D. All of these

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Bacteria reproduce primarily by binary fission, form spores under stress, and exchange genetic material through conjugation and other mechanisms, making all statements correct.

QUESTION 120 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · MCQ · 4 marks · 7s

Heterotrophic bacteria help in:

OPTIONS

A. Curdling of milk.

B. Production of antibiotics.

C. Nitrogen fixation in leguminous plants.

D. All of these

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Heterotrophic bacteria perform diverse functions including curdling milk (Lactobacillus), producing antibiotics (Streptomyces), and nitrogen fixation (Rhizobium), making all options correct.

QUESTION 121 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · MCQ · 4 marks · 1m 14s

Select the total number of diseases caused by bacteria from the following: Mumps, Smallpox, Citrus canker, Cholera, Typhoid, Tetanus, Sleeping sickness, Malaria.

OPTIONS

A. 2

B. 4

C. 5

D. 6

STUDENT RESPONSE
B

CORRECT ANSWER
B

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Four diseases (Citrus canker, Cholera, Typhoid, Tetanus) are caused by bacteria, while Mumps and Smallpox are viral, and Sleeping sickness and Malaria are protozoan.

QUESTION 122 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · MATCH_COLUMN · 4 marks · 2m 43s

Match List-I with List-II. List-I: a. Chemosynthetic bacteria, b. Cyanobacteria, c. Heterotrophic bacteria, d. Mycoplasma. List-II: i. Majority are important decomposers, ii. Release oxygen, iii. Oxidise nitrites, iv. Smallest living cells.

OPTIONS

A. a-i, b-ii, c-iii, d-iv

B. a-ii, b-iii, c-iv, d-i

C. a-iv, b-ii, c-i, d-iii

D. a-iii, b-ii, c-i, d-iv

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Chemosynthetic bacteria oxidize nitrites; cyanobacteria release oxygen through photosynthesis; heterotrophic bacteria are mainly decomposers; and mycoplasma are the smallest living cells.

QUESTION 123 BIOLOGY

CORRECT

Biological Classification · Monera · Medium · ASSERTION_REASON · 4 marks · 14s

Statement-I: Eubacteria are characterised by the presence of a rigid cell wall. Statement-II: Cyanobacteria often form blooms in polluted water bodies.

OPTIONS

A. Statement I and Statement II both are correct.

B. Statement I is correct, but Statement II is incorrect.

C. Statement I is incorrect, but Statement II is correct.

D. Statement I and Statement II both are incorrect.

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Eubacteria have peptidoglycan-based rigid cell walls, and cyanobacteria commonly form blooms in nutrient-polluted waters due to their rapid growth in eutrophic conditions.

QUESTION 124 BIOLOGY

INCORRECT

Biological Classification · Monera · Medium · ASSERTION_REASON · −1 mark · 13s

Assertion (A): Bacteria are pathogens causing damage to human beings, crops, farm animals and pets. Reason (R): Citrus canker affects plants.

OPTIONS

A. Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).

B. Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).

C. Assertion (A) is true and Reason (R) is false.

D. Assertion (A) is false and Reason (R) is true.

STUDENT RESPONSE
A

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Conceptual Error · Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

EXPLANATION

Both statements are factually correct, but Reason (R) citing only one plant disease cannot explain the broad assertion about bacteria affecting humans, animals, and pets.

MISTAKE NOTE

Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

QUESTION 125 BIOLOGY

CORRECT

Biological Classification · Protista · Medium · MCQ · 4 marks · 16s

What is NOT true about Euglena?

OPTIONS

A. Presence of cellulosic cell wall.

B. Presence of proteinaceous pellicle.

C. Presence of chlorophyll.

D. Presence of flagellum.

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Euglena lacks a cellulosic cell wall and instead has a proteinaceous pellicle, making option (a) the incorrect statement.

QUESTION 126 BIOLOGY

CORRECT

Biological Classification · Protista · Medium · MCQ · 4 marks · 31s

Planktons are those organisms which:

OPTIONS

A. Float actively in water current.

B. Swim passively in water current.

C. Float passively in air current.

D. Float passively in water current.

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Plankton are microscopic organisms that drift or float passively in water bodies, unable to swim against currents. They include phytoplankton and zooplankton, and are exclusively aquatic, not airborne.

QUESTION 127 BIOLOGY

INCORRECT

Biological Classification · Protista · Medium · MCQ · −1 mark · 27s

Diatomaceous earth is formed after (A) of years because the walls are embedded with (B) and thus are (C). Identify A, B and C.

OPTIONS

A. Millions, cellulose, destructible

B. Billions, cellulose, indestructible

C. Millions, silica, indestructible

D. Billions, silica, indestructible

STUDENT RESPONSE
D

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Conceptual Error · Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

EXPLANATION

Diatomaceous earth forms from accumulation of dead diatoms over millions of years; their cell walls contain silica, making them indestructible and able to form sedimentary deposits.

MISTAKE NOTE

Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

QUESTION 128 BIOLOGY

CORRECT

Biology · Medium · 4 marks · 4 marks · 57s

Dinoflagellates have: groove between the cell plates. longitudinal groove between the cell plates. and the other transversely in a furrow between the wall plates.

OPTIONS

A. Two flagella, which lie longitudinally.

B. Only one flagellum in the transverse

C. Only one flagellum in the

D. One flagellum lying longitudinally

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Dinoflagellates possess two distinct flagella: one transverse flagellum in the cingulum (causing rotation) and one longitudinal flagellum in the sulcus (acting as rudder), both located in grooves between thecal plates.

QUESTION 129 BIOLOGY

CORRECT

Biological Classification · Protista · Medium · MCQ · 4 marks · 11s

Which of the following protists releases toxins that may even kill fishes and other marine animals?

OPTIONS

A. *Euglena*

B. *Gonyaulax*

C. Diatoms

D. Slime moulds

STUDENT RESPONSE
B

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Gonyaulax is a dinoflagellate that causes 'red tides' and produces powerful neurotoxins (saxitoxin) that kill fish and other marine animals, and can cause paralytic shellfish poisoning in humans.

QUESTION 130 BIOLOGY

CORRECT

Biological Classification · Protista · Medium · MCQ · 4 marks · 30s

Primitive relatives of animals are:

OPTIONS

A. Euglenoids

B. Slime moulds

C. Protozoans

D. Dinoflagellates

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Protozoans are unicellular, heterotrophic protists lacking cell walls, making them the most primitive relatives of animals in the evolutionary lineage.

QUESTION 131 BIOLOGY

CORRECT

Biological Classification · Protista · Medium · MCQ · 4 marks · 13s

is an aquatic and actively moving organism due to the presence of:

OPTIONS

A. Pseudopodia

B. False feet

C. Thousands of cilia

D. Flagella

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Paramecium possesses thousands of cilia covering its body surface that beat rhythmically to propel the organism through water and also help in directing food into the oral groove.

QUESTION 132 BIOLOGY

CORRECT

Biological Classification · Protista · Medium · MCQ · 4 marks · 10s

Pick the odd one out w.r.t. uses of diatomaceous earth.

OPTIONS

A. Biogas production and degradation

B. Filtration of oils

C. Polishing

D. Filtration of syrups

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Biogas production and degradation is not a standard use of diatomaceous earth, whereas filtration of oils/syrups and polishing are well-established applications.

QUESTION 133 BIOLOGY

CORRECT

Human Health and Diseases · Malaria · Medium · MCQ · 4 marks · 8s

Sleeping sickness is caused by:

OPTIONS

A. *Entamoeba*

B. *Plasmodium*

C. Slime moulds

D. *Trypanosoma*

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Trypanosoma brucei is the protozoan parasite that causes African sleeping sickness, transmitted through tsetse fly bites.

QUESTION 134 BIOLOGY

CORRECT

Biological Classification · Protista · Medium · MCQ · 4 marks · 6s

Infectious spore-like stage in the life cycle is found in:

OPTIONS

A. Amoeboid protozoans

B. Ciliated protozoans

C. Flagellated protozoans

D. Sporozoans

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Sporozoans are characterized by the presence of spore-like infectious stages called sporozoites in their life cycle, which help in transmission to new hosts. This is a defining feature of this class of protozoans.

QUESTION 135 BIOLOGY

INCORRECT

Biological Classification · Protista · Medium · MCQ · −1 mark · 9s

The spores produced in slime moulds are dispersed by:

OPTIONS

A. Water

B. Air currents

C. Birds

D. Both (1) and (2)

STUDENT RESPONSE
D

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Conceptual Error · Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

EXPLANATION

Slime moulds produce spores in sporangia that are lightweight and primarily dispersed by air currents to colonize new habitats.

MISTAKE NOTE

Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

QUESTION 136 BIOLOGY

CORRECT

Animal Kingdom · Basis of Classification · Medium · MCQ · 4 marks · 18s

In some animal groups, the body is found divided into compartments with serial repetition of at least some organs. This characteristic feature is named as:

OPTIONS

A. Locomotion

B. Metamerism

C. Metagenesis

D. Metamorphosis

STUDENT RESPONSE

B

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Metamerism is the characteristic feature where body is divided into segments (metameres) with serial repetition of organs, seen in annelids, arthropods, and chordates.

QUESTION 137 BIOLOGY

CORRECT

Animal Kingdom · Basis of Classification · Medium · MCQ · 4 marks · 9s

Notochord is derived from:

OPTIONS

A. Mesoderm layer

B. Ectoderm layer

C. Endoderm layer

D. None of these

STUDENT RESPONSE

A

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

The notochord is derived from the mesoderm layer, specifically from the notochordal plate formed by mesodermal cells migrating through the primitive streak during embryonic development.

QUESTION 138 BIOLOGY

CORRECT

Animal Kingdom · Basis of Classification · Medium · MCQ · 4 marks · 18s

Body cavity is the cavity present between body wall and gut wall. In some animals, the body cavity is not lined by mesoderm, instead, the mesoderm is present scattered as pouches in between the ectoderm and endoderm. This is called:

OPTIONS

A. Acoelomate

B. Coelomate

C. Pseudocoelomate

D. Both (1) and (2)

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Pseudocoelomates have a body cavity (pseudocoel) that is not lined by mesoderm; instead mesoderm forms scattered pouches between ectoderm and endoderm, with the cavity being the blastocoel persisting between endoderm and mesoderm.

QUESTION 139 BIOLOGY

CORRECT

Animal Kingdom · Coelenterata · Medium · MCQ · 4 marks · 8s

belongs to the phylum:

OPTIONS

A. Porifera

B. Coelenterata

C. Ctenophora

D. Platyhelminthes

STUDENT RESPONSE
B

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Aurelia (moon jellyfish) is a cnidarian belonging to phylum Coelenterata (Cnidaria), characterized by radial symmetry, tentacles with stinging cells (cnidocytes), and a gelatinous body.

QUESTION 140 BIOLOGY

CORRECT

Animal Kingdom · Porifera · Medium · MCQ · 4 marks · 7s

Water → Ostia → Spongocoel → Osculum. The above pathway of water helps in:

OPTIONS

A. Food gathering

B. Respiratory exchange

C. Removal of waste

D. All of these

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION

The canal system in sponges serves multiple functions simultaneously - food capture through filter feeding, gas exchange for respiration, and excretion of waste products, all facilitated by the continuous water flow.

QUESTION 141 BIOLOGY

CORRECT

Animal Kingdom · Porifera · Medium · MCQ · 4 marks · 25s

Select the incorrect statement about poriferans.

OPTIONS

A. They are commonly known as sponges.

B. They are sessile, solitary or colonial.

C. Mostly asymmetrical, rarely radially symmetrical (e.g., $Sycon$).

D. Single ostium and many oscula are present in them.

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION

Poriferans have many ostia (incurrent pores) for water entry and one or few oscula (excurrent openings) for water exit, not the reverse as stated in option (d).

QUESTION 142 BIOLOGY

CORRECT

Animal Kingdom · Coelenterata · Medium · MCQ · 4 marks · 14s

I. Tissue level of body organisation II. Diploblastic III. Gastrovascular cavity present IV. Exhibit alternation of generation. The above characteristics belong to the phylum:

OPTIONS

A. Cnidaria

B. Porifera

C. Platyhelminthes

D. Ctenophora

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Cnidaria are the only phylum among the options that are diploblastic, have tissue-level organization, possess a gastrovascular cavity, and exhibit alternation of generations (metagenesis) between polyp and medusa forms.

QUESTION 143 BIOLOGY

CORRECT

Animal Kingdom · Porifera · Medium · MCQ · 4 marks · 11s

Select the incorrectly matched pair regarding phylum Porifera.

OPTIONS

A. Choanocytes – collar cells

B. Inner choanoderm – contains choanocytes

C. $Sycon$ – $Scypha$

D. $Sycon$ – bath sponge

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Sycon is a small calcareous sponge, while bath sponge refers to large commercial demosponges like Spongia. The bath sponge is incorrectly matched with Sycon.

QUESTION 144 BIOLOGY

CORRECT

Animal Kingdom · Coelenterata · Medium · MCQ · 4 marks · 9s

Which of the following is/are true about the cnidoblasts/cnidocytes?

OPTIONS

A. They are present on the tentacles and the body.

B. They are used for anchorage, defense and for the capture of prey.

C. They contain stinging capsules or nematocysts.

D. All of these

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Cnidocytes are characteristic cells of Cnidaria containing nematocysts, found throughout the body and tentacles, serving multiple functions including prey capture, defense, and attachment.

QUESTION 145 BIOLOGY

INCORRECT

Animal Kingdom · Porifera · Medium · MCQ · −1 mark · 27s

Which type of fertilisation and development is seen in poriferans?

OPTIONS

A. Internal and direct

B. Internal and indirect

C. External and direct

D. External and indirect

STUDENT RESPONSE
B

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Conceptual Error · Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

EXPLANATION

Poriferans exhibit external fertilization as sperm is released into water to reach eggs, and indirect development due to presence of free-swimming larval stage before metamorphosis into adult.

MISTAKE NOTE

Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

QUESTION 146 BIOLOGY

CORRECT

Animal Kingdom · Porifera · Medium · MCQ · 4 marks · 15s

Which of the following is a unique characteristic of pore bearing animals?

OPTIONS

A. Choanocytes or collar cells are present in them that line the spongocoel and the canals.

B. They are hermaphrodite.

C. They respire through general body surface.

D. They reproduce by asexual means only.

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Choanocytes or collar cells are the defining unique feature of Porifera (pore-bearing animals), as these specialized flagellated cells for filter feeding are not found in any other animal phylum.

QUESTION 147 BIOLOGY

CORRECT

Animal Kingdom · Porifera · Medium · MCQ · 4 marks · 10s

Which of the following is a fresh water sponge?

OPTIONS

A. *Physalia*

B. *Euspongia*

C. *Spongilla*

D. *Sycon*

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Spongilla is the only freshwater sponge among the options; all others are marine organisms.

QUESTION 148 BIOLOGY

CORRECT

Animal Kingdom · Coelenterata · Medium · MCQ · 4 marks · 6s

Those cnidarians which exist in both forms (polyp and medusa) exhibit:

OPTIONS

A. Metamerism

B. Metagenesis

C. Metamorphosis

D. All of these

STUDENT RESPONSE
B

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Metagenesis is the phenomenon of alternation of generations between asexual polyp stage and sexual medusa stage in cnidarians like Obelia.

QUESTION 149 BIOLOGY

CORRECT

Animal Kingdom · Coelenterata · Medium · MCQ · 4 marks · 18s

Statement-I: Some cnidarians, e.g., corals have a skeleton composed of spongin fibre. Statement-II: They have a central gastro-vascular cavity with a single opening, mouth, on hypostome.

OPTIONS

A. Statement I and Statement II both are correct.

B. Statement I is correct, but Statement II is incorrect.

C. Statement I is incorrect, but Statement II is correct.

D. Statement I and Statement II both are incorrect.

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Corals have calcium carbonate skeletons, not spongin (which is found in sponges); however, cnidarians do possess a gastro-vascular cavity with single opening on hypostome.

QUESTION 150 BIOLOGY

CORRECT

Animal Kingdom · Porifera · Medium · ASSERTION_REASON · 4 marks · 19s

Assertion (A): Cellular level of body organisation is found in poriferans. Reason (R): Poriferans' bodies are composed of specialised cells that perform specific functions, but these cells are not organised into tissues or organs like in more complex organisms.

OPTIONS

A. Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).

B. Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).

C. Assertion (A) is true, and Reason (R) is false.

D. Assertion (A) is false, and Reason (R) is true.

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Poriferans exhibit cellular level of organisation because their specialized cells (choanocytes, pinacocytes, amoebocytes) function independently without forming true tissues or organs, making option (a) correct.

QUESTION 151 BIOLOGY

CORRECT

Animal Kingdom · Ctenophora · Easy · MCQ · 4 marks · 11s

Ctenophores are commonly known as:

OPTIONS

A. Flatworms

B. Sponges

C. Sea walnuts or comb jellies

D. Tapeworms

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Ctenophores are marine invertebrates with distinctive comb-like ciliary plates, earning them the common names sea walnuts or comb jellies.

QUESTION 152 BIOLOGY

CORRECT

Animal Kingdom · Platyhelminthes · Medium · MCQ · 4 marks · 11s

Specialised cells in Platyhelminthes called ___ help in osmoregulation and excretion.

OPTIONS

A. Choanocytes

B. Cnidocytes

C. Nerve cells

D. Flame cells

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Flame cells are specialised excretory cells in Platyhelminthes that regulate water balance and remove metabolic wastes through a network of tubules opening to the exterior.

QUESTION 153 BIOLOGY

CORRECT

Animal Kingdom · Platyhelminthes · Medium · MCQ · 4 marks · 11s

Which of the following phyla shows bilateral symmetry, triploblastic and acoelomate characteristics?

OPTIONS

A. Platyhelminthes

B. Cnidaria

C. Porifera

D. Ctenophora

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Platyhelminthes (flatworms) are the simplest triploblastic animals with bilateral symmetry and no body cavity (acoelomate), making them the correct answer.

QUESTION 154 BIOLOGY

CORRECT

Animal Kingdom · Platyhelminthes · Medium · MCQ · 4 marks · 7s

Which of the following organisms belongs to Platyhelminthes and also possesses high regeneration capacity?

OPTIONS

A. *Sycon*

B. *Planaria*

C. *Adamsia*

D. *Ctenoplana*

STUDENT RESPONSE

B

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Planaria is a free-living flatworm belonging to phylum Platyhelminthes and exhibits extraordinary regeneration capacity, able to regenerate entire organism from a small body fragment.

QUESTION 155 BIOLOGY

CORRECT

Animal Kingdom · Platyhelminthes · Easy · MCQ · 4 marks · 7s

Flame cells are the characteristic feature of phylum:

OPTIONS

A. Platyhelminthes

B. Ctenophora

C. Porifera

D. Coelenterata

STUDENT RESPONSE

A

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Flame cells, also called protonephridia, are the characteristic excretory organs of Platyhelminthes (flatworms), functioning in osmoregulation and waste removal.

QUESTION 156 BIOLOGY

CORRECT

Animal Kingdom · Ctenophora · Medium · MCQ · 4 marks · 1m 15s

Select the correct statement(s) about phylum Ctenophora.

OPTIONS

A. Sexes are separate.

B. Fertilisation is external.

C. Development is indirect.

D. Both (2) and (3)

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Ctenophora are hermaphroditic (not separate sexes), but exhibit external fertilization and indirect development with cydippid larva. Since statements (b) and (c) are both correct, option (d) is the answer.

QUESTION 157 BIOLOGY

CORRECT

Animal Kingdom · Ctenophora · Medium · MCQ · 4 marks · 7s

The body bears (A) external rows of ciliated comb plates, which help in (B).

OPTIONS

A. A – six, B – locomotion

B. A – six, B – respiration

C. A – eight, B – locomotion

D. A – eight, B – respiration

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Ctenophores (comb jellies) have eight meridional rows of ciliated comb plates that beat synchronously to propel the animal through water, serving as the primary mode of locomotion.

QUESTION 158 BIOLOGY

CORRECT

Animal Kingdom · Platyhelminthes · Medium · MCQ · 4 marks · 9s

Select the incorrect statement about phylum Platyhelminthes.

OPTIONS

A. These are mostly endoparasites found in animals including human beings.

B. Hooks and suckers are present in the parasitic forms.

C. Specialised cells called flame cells help in osmoregulation and excretion.

D. *Planaria* possesses low regeneration capacity.

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Planaria is well-known for its remarkable high regeneration capacity, making statement (d) incorrect.

QUESTION 159 BIOLOGY

CORRECT

Animal Kingdom · Ctenophora · Easy · MCQ · 4 marks · 10s

Ctenoplana and Pleurobrachia are examples of phylum:

OPTIONS

A. Porifera

B. Cnidaria

C. Platyhelminthes

D. Ctenophora

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION

Ctenoplana and Pleurobrachia are marine comb jellies that possess characteristic ciliary comb plates for swimming, placing them in phylum Ctenophora.

QUESTION 160 BIOLOGY

SKIPPED

Animal Kingdom · Aschelminthes · Medium · MCQ · 0 marks · 3m 57s

Female Ascaris can be identified on the basis of:

OPTIONS

A. Presence of two spicules found at the posterior end.

B. Presence of post-anal and pre-anal papillae.

C. Presence of straight posterior end.

D. Presence of common cloacal aperture.

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
C

PERFORMANCE INSIGHT
Not attempted — no marks awarded

EXPLANATION

Female Ascaris can be distinguished from males by its larger size, straight/blunt posterior end, and separate anal and genital openings, while males have a curved posterior end with spicules and papillae for copulation.

QUESTION 161 BIOLOGY

CORRECT

Animal Kingdom · Aschelminthes · Medium · MCQ · 4 marks · 29s

Roundworms represent an example of:

OPTIONS

A. Cellular grade of organisation

B. Tissue grade of organisation

C. Organ-system grade of organisation

D. Tissue-organ grade of organisation

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Roundworms (Nematoda) exhibit organ-system grade of organization with well-developed digestive, reproductive, nervous, and excretory systems functioning in coordination.

QUESTION 162 BIOLOGY

SKIPPED

Animal Kingdom · Aschelminthes · Easy · MCQ · 0 marks · 1m 43s

is a parasite of:

OPTIONS

A. Large intestine

B. Small intestine

C. Liver

D. Stomach

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Ascaris lumbricoides, the human intestinal roundworm, lives as an adult parasite in the lumen of the small intestine where it feeds on digested nutrients.

QUESTION 163 BIOLOGY

CORRECT

Animal Kingdom · Aschelminthes · Hard · MCQ · 4 marks · 43s

Consider the following statements about Aschelminthes. (I) Their body is circular in cross-section, so are called roundworms. (II) Alimentary canal is incomplete. (III) Muscular pharynx is present. (IV) All members are hermaphrodites. Which option contains all the correct statement(s)?

OPTIONS

A. I and III

B. II and IV

C. I, II and IV

D. IV only

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Aschelminthes are roundworms with circular cross-section, complete digestive tract with muscular pharynx, and are mostly dioecious, not hermaphroditic.

QUESTION 164 BIOLOGY

CORRECT

Animal Kingdom · Annelida · Medium · MCQ · 4 marks · 17s

Which of the following animals are true coelomates with bilateral symmetry?

OPTIONS

A. Adult echinoderms

B. Aschelminthes

C. Platyhelminthes

D. Annelids

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Annelids possess a true coelom (schizocoelom lined by mesoderm) and maintain bilateral symmetry, unlike echinoderms (radial symmetry as adults), Aschelminthes (pseudocoelomates), and Platyhelminthes (acoelomates).

QUESTION 165 BIOLOGY

INCORRECT

Biology · Medium · −1 mark · −1 mark · 32s

In annelids: ganglia connected by lateral nerves to a double ventral nerve cord. and sexually. Hirudinaria are monoecious. absent.

OPTIONS

A. The neural system consists of paired

B. Reproduction occurs both asexually

C. Like Nereis, Pheretima and

D. Aquatic forms are completely

STUDENT RESPONSE
A

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Conceptual Error · Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

EXPLANATION

Annelids exhibit both asexual reproduction (fragmentation, budding, fission) and sexual reproduction (most are hermaphroditic). Option (c) is wrong because Nereis is dioecious, not monoecious. Option (d) is wrong as many polychaetes are aquatic/marine.

MISTAKE NOTE

Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

QUESTION 166 BIOLOGY

CORRECT

Animal Kingdom · Annelida · Medium · MCQ · 4 marks · 15s

Which of the following statements is incorrect about annelids?

OPTIONS

A. Also known as roundworms.

B. Their body surface is distinctly marked out into segments or metameres.

C. Nephridia help in osmoregulation and excretion.

D. They may be aquatic (marine and fresh water) or terrestrial.

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Annelids are segmented worms, not roundworms. Roundworms belong to phylum Nematoda, while annelids belong to phylum Annelida. The term 'roundworms' specifically refers to nematodes, making statement (a) incorrect.

QUESTION 167 BIOLOGY

CORRECT

Animal Kingdom · Aschelminthes · Easy · MCQ · 4 marks · 9s

Aschelminthes are called roundworms because:

OPTIONS

A. Their body is round like a disc.

B. Their thread-like body is circular in cross section.

C. Their stomach is round shaped.

D. Their larvae are round in shape.

STUDENT RESPONSE
B

CORRECT ANSWER
B

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Aschelminthes are called roundworms because their elongated, thread-like body appears circular in cross-section, not flat or ribbon-like.

QUESTION 168 BIOLOGY

CORRECT

Animal Kingdom · Aschelminthes · Easy · MCQ · 4 marks · 9s

Aschelminthes may be:

OPTIONS

A. Free-living

B. Aquatic and terrestrial

C. Parasitic in animals

D. All of these

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Aschelminthes exhibit diverse lifestyles including free-living forms in aquatic and terrestrial habitats, as well as parasitic forms in animals, making all options correct.

QUESTION 169 BIOLOGY

INCORRECT

Animal Kingdom · Arthropoda · Medium · MCQ · −1 mark · 10s

Which of the following is/are an exclusive characteristic(s) of the members of phylum Arthropoda?

OPTIONS

A. They have jointed appendages.

B. Excretion takes place through Malpighian tubules.

C. They show organ-system level of organisation.

D. All of these

STUDENT RESPONSE
D

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Conceptual Error · Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

EXPLANATION

Jointed appendages are the unique, diagnostic feature of phylum Arthropoda (giving them their name). Malpighian tubules are not present in all arthropods (crustaceans use antennal glands), and organ-system level organization is shared with many other phyla.

MISTAKE NOTE

Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

QUESTION 170 BIOLOGY

CORRECT

Animal Kingdom · Arthropoda · Medium · MCQ · 4 marks · 5s

The respiratory organs of arthropods are:

OPTIONS

A. Book gills

B. Book lungs

C. Tracheal system

D. All of these

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Arthropods exhibit diverse respiratory adaptations: book gills in aquatic forms (e.g., Limulus), book lungs in terrestrial arachnids (e.g., spiders), and tracheal system in insects, making all three valid respiratory organs of this phylum.

QUESTION 171 BIOLOGY

CORRECT

Animal Kingdom · Arthropoda · Easy · MCQ · 4 marks · 10s

Arthropods have sense organs like:

OPTIONS

A. Antennae

B. Statocysts for balance

C. Simple and compound eyes

D. All of these

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Arthropods possess diverse sense organs including antennae for chemoreception and mechanoreception, statocysts for equilibrium detection, and both simple and compound eyes for vision, making all options correct.

QUESTION 172 BIOLOGY

CORRECT

Animal Kingdom · Arthropoda · Easy · MCQ · 4 marks · 8s

Economically important insect(s) is/are:

OPTIONS

A. *Apis*

B. *Laccifer*

C. *Bombyx*

D. All of these

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Apis (honey bee), Laccifer (lac insect), and Bombyx (silkworm) are all economically important insects producing honey, lac, and silk respectively.

QUESTION 173 BIOLOGY

CORRECT

Animal Kingdom · Arthropoda · Hard · MCQ · 4 marks · 13s

Select the incorrect pair w.r.t. arthropods.

OPTIONS

A. Haemolymph – Insect

B. Haemocyanin – Prawn

C. Germ layer – Diploblastic

D. Circulation – Open circulatory system

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION
Arthropods are triploblastic animals with three germ layers, not diploblastic which only have two germ layers.

QUESTION 174 BIOLOGY

CORRECT

Animal Kingdom · Arthropoda · Medium · MCQ · 4 marks · 7s

Select the correct statement.

OPTIONS

A. Terrestrial arthropods excrete ammonia.

B. Aquatic arthropods excrete uric acid.

C. Arthropods have jointed appendages.

D. All named species on earth are arthropods.

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Arthropods are defined by their jointed appendages, which is the characteristic feature from which their name is derived. Options (a) and (b) are reversed in their excretory patterns, and (d) is factually incorrect as arthropods represent only a majority, not all species.

QUESTION 175 BIOLOGY

CORRECT

Animal Kingdom · Arthropoda · Medium · MCQ · 4 marks · 12s

Select the incorrect statement.

OPTIONS

A. Arthropods have tissue level of body organisation.

B. Arthropods are bilaterally symmetrical.

C. Body of arthropods consists of head, thorax and abdomen.

D. Mostly arthropods are dioecious.

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Arthropods have organ-system level of organization, not tissue level. Tissue level organization is characteristic of lower phyla like Cnidaria and Platyhelminthes.

QUESTION 176 BIOLOGY

CORRECT

Animal Kingdom · Arthropoda · Medium · MCQ · 4 marks · 7s

Select the odd one out.

OPTIONS

A. *Anopheles*

B. *Limulus*

C. *Culex*

D. *Aedes*

STUDENT RESPONSE
B

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Anopheles, Culex, and Aedes are all mosquito genera (Insecta), while Limulus (horseshoe crab) belongs to class Merostomata, making it taxonomically distinct.

QUESTION 177 BIOLOGY

CORRECT

Animal Kingdom · Arthropoda · Easy · MCQ · 4 marks · 4s

Which of the following is an economically important insect?

OPTIONS

A. *Bombyx*

B. *Aedes*

C. *Anopheles*

D. *Culex*

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Bombyx mori (silkworm) is the source of natural silk, making it economically important through sericulture, while the others are disease-carrying mosquitoes.

QUESTION 178 BIOLOGY

INCORRECT

Animal Kingdom · Arthropoda · Medium · MCQ · −1 mark · 26s

Arthropods are (A)___ and (B)___.

OPTIONS

A. A – diploblastic, B – segmented

B. A – diploblastic, B – unsegmented

C. A – triploblastic, B – segmented

D. A – triploblastic, B – unsegmented

STUDENT RESPONSE
D

CORRECT ANSWER
C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Conceptual Error · Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

EXPLANATION

Arthropods are triploblastic (three germ layers: ectoderm, mesoderm, endoderm) and have a segmented body with jointed appendages, which are defining characteristics of the phylum Arthropoda.

MISTAKE NOTE

Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

QUESTION 179 BIOLOGY

CORRECT

Animal Kingdom · Arthropoda · Medium · MCQ · 4 marks · 13s

Blood flows in open tissue spaces and ___ instead of blood vessels.

OPTIONS

A. Pseudocoel

B. Hemocoel

C. Gills

D. Lungs

STUDENT RESPONSE
B

CORRECT ANSWER
B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Hemocoel is the body cavity in arthropods and some mollusks where hemolymph (blood) flows freely in open spaces surrounding organs, characteristic of an open circulatory system.

Biology · Q180 · question continued

QUESTION 180 BIOLOGY

CORRECT

Animal Kingdom · Echinodermata · Medium · MCQ · 4 marks · 25s

Which of the following statements represents the incorrect feature of echinoderms?

OPTIONS

A. They are triploblastic and coelomate animals.

B. All are marine with cellular level of organisation.

C. Endoskeleton of calcareous ossicles.

D. None of these

STUDENT RESPONSE

B

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Echinoderms have organ system level of organisation, not cellular level. Cellular level organisation is characteristic of Porifera (sponges). 06 LEARNING INTELLIGENCE Learning intelligence Published revision priorities Priority revision register PRIORITY TOPIC CLASSIFICATION TIMEFRAME Position and Displacement High This week Scalars and Vectors High This week Average and Instantaneous Velocity High This week Uniform Circular Motion High This week Topic-level accuracy TOPIC SUBJECT ACCURACY LEVEL Position and Displacement Physics 42.9% Weak Scalars and Vectors Physics 40% Weak Average and Instantaneous Velocity Physics 41.7% Weak Uniform Circular Motion Physics 0% Critical Equations of Motion Physics 33.3% Critical Acceleration Physics 0% Critical Relative Velocity in 2D Physics 0% Critical Work-Energy Theorem Physics 100% Strong Vector Addition Physics 0% Critical Physics Physics 0% Critical Dalton's Atomic Theory Chemistry 100% Strong Rutherford Model Chemistry 66.7% Average Thomson Model Chemistry 40% Weak Chemistry Chemistry 66.7% Average Isotopes Chemistry 66.7% Average Isobars Chemistry 100% Strong Properties of EM Waves Chemistry 100% Strong de Broglie Relation Chemistry 0% Critical Topic-level accuracy · continued TOPIC SUBJECT ACCURACY LEVEL Electronic Configuration Chemistry 100% Strong Quantum Numbers Chemistry 50% Weak Mole Concept Chemistry 7.1% Critical Five Kingdom Classification Biology 81.8% Strong Monera Biology 82.6% Strong Protista Biology 77.8% Strong Biology Biology 50% Weak Malaria Biology 100% Strong Basis of Classification Biology 100% Strong Coelenterata Biology 100% Strong Porifera Biology 85.7% Strong Ctenophora Biology 100% Strong Platyhelminthes Biology 100% Strong Aschelminthes Biology 66.7% Average Annelida Biology 100% Strong Arthropoda Biology 81.8% Strong Echinodermata Biology 100% Strong Mistake evidence QUESTION SOURCE DETAIL Q3 Calculation Error · Position and Displacement Q4 Calculation Error · Position and Displacement Q10 Knowledge Gap · Average and Instantaneous Velocity Q11 Knowledge Gap · Average and Instantaneous Velocity Q12 Knowledge Gap · Average and Instantaneous Velocity Q15 Knowledge Gap · Average and Instantaneous Velocity Q22 Calculation Error · Acceleration Q26 Knowledge Gap · Acceleration Q33 Formula Error · Equations of Motion Q35 Formula Error · Scalars and Vectors Mistake evidence · continued QUESTION SOURCE DETAIL Q37 Knowledge Gap · Scalars and Vectors Q43 Calculation Error · Vector Addition Q44 Knowledge Gap · Physics Q59 Calculation Error · Rutherford Model Q61 Calculation Error · Rutherford Model Q64 Knowledge Gap · Thomson Model Q68 Knowledge Gap · Isotopes Q73 Formula Error · de Broglie Relation Q77 Knowledge Gap · Mole Concept Q85 Knowledge Gap · Mole Concept Q87 Knowledge Gap · Mole Concept Q89 Knowledge Gap · Mole Concept Q92 Knowledge Gap · Five Kingdom Classification Q98 Knowledge Gap · Five Kingdom Classification Q112 Conceptual Error · Monera Q113

Biology · Q180 · explanation continued

Conceptual Error · Monera Q115 Knowledge Gap · Monera Q124 Conceptual Error · Monera Q127 Conceptual Error · Protista Q135 Conceptual Error · Protista Q145 Conceptual Error · Porifera Q165 Conceptual Error · Biology Q169 Conceptual Error · Arthropoda Q178 Conceptual Error · Arthropoda Weak chapters NO. WEAK CHAPTER 1 Motion in a Straight Line 2 Motion in a Plane 3 Physics Weak chapters · continued NO. WEAK CHAPTER 4 Structure of Atom 5 Some Basic Concepts of Chemistry 6 Biology Subject-wise revision sections Priority topics grouped by subject PRIORITY TOPIC CHAPTER ACCURACY CLASSIFICATION Physics Uniform Circular Motion Motion in a Plane 0% Critical Acceleration Motion in a Straight Line 0% Critical Relative Velocity in 2D Motion in a Plane 0% Critical Vector Addition Motion in a Plane 0% Critical Equations of Motion Motion in a Straight Line 33.3% Critical Scalars and Vectors Motion in a Plane 40% Critical Average and Instantaneous Velocity Motion in a Straight Line 41.7% Critical Position and Displacement Motion in a Straight Line 42.9% Critical Chemistry de Broglie Relation Structure of Atom 0% Critical Mole Concept Some Basic Concepts of Chemistry 7.1% Critical Thomson Model Structure of Atom 40% Critical Quantum Numbers Structure of Atom 50% Weak Recommended next actions Weak topics · published list NO. WEAK TOPIC 1 Position and Displacement 2 Scalars and Vectors 3 Average and Instantaneous Velocity 4 Uniform Circular Motion 5 Equations of Motion 6 Acceleration Weak topics · published list · continued NO. WEAK TOPIC 7 Relative Velocity in 2D 8 Vector Addition 9 Physics 10 Thomson Model 11 de Broglie Relation 12 Quantum Numbers 13 Mole Concept 14 Biology MediXO EduX · Examination Intelligence End of report · 03 Nov 2026

05 LEARNING INTELLIGENCE

Learning intelligence

Published revision priorities

Mistake evidence

QUESTION	SOURCE DETAIL
Q3	Calculation Error · Position and Displacement
Q4	Calculation Error · Position and Displacement
Q10	Knowledge Gap · Average and Instantaneous Velocity
Q11	Knowledge Gap · Average and Instantaneous Velocity
Q12	Knowledge Gap · Average and Instantaneous Velocity
Q15	Knowledge Gap · Average and Instantaneous Velocity
Q22	Calculation Error · Acceleration
Q26	Knowledge Gap · Acceleration
Q33	Formula Error · Equations of Motion
Q35	Formula Error · Scalars and Vectors
Q37	Knowledge Gap · Scalars and Vectors

Mistake evidence · continued

QUESTION	SOURCE DETAIL
Q43	Calculation Error · Vector Addition
Q44	Knowledge Gap · Medium
Q59	Calculation Error · Rutherford Model
Q61	Calculation Error · Rutherford Model
Q64	Knowledge Gap · Thomson Model
Q68	Knowledge Gap · Isotopes
Q73	Formula Error · de Broglie Relation
Q77	Knowledge Gap · Mole Concept
Q85	Knowledge Gap · Mole Concept
Q87	Knowledge Gap · Mole Concept
Q89	Knowledge Gap · Mole Concept
Q92	Knowledge Gap · Five Kingdom Classification
Q98	Knowledge Gap · Five Kingdom Classification
Q112	Conceptual Error · Monera
Q113	Conceptual Error · Monera
Q115	Knowledge Gap · Monera
Q124	Conceptual Error · Monera
Q127	Conceptual Error · Protista
Q135	Conceptual Error · Protista
Q145	Conceptual Error · Porifera
Q165	Conceptual Error · Medium
Q169	Conceptual Error · Arthropoda
Q178	Conceptual Error · Arthropoda

Subject-wise revision sections

Recommended next actions